
FOLIO SUBMISSION

Kankawee Maksomboon | s4097770

ELECTIVE BASE RESEARCH ASSISTANT

Tectonic formation lab

Marc Gibson

Research team : Charles Boman | Lachlan Buckley
Fabrication team : Tiffany Vuong | Ansel Lin | Kory Ko | Takdanai Phakdee



The background of the entire page is a composite image. The upper portion features a close-up of a sculpture made of a light-colored, textured material, possibly plaster or concrete, showing the profile of a face and a hand. The lower portion features a topographical model made of thin, parallel lines, creating a sense of depth and relief.

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INTRODUCTION

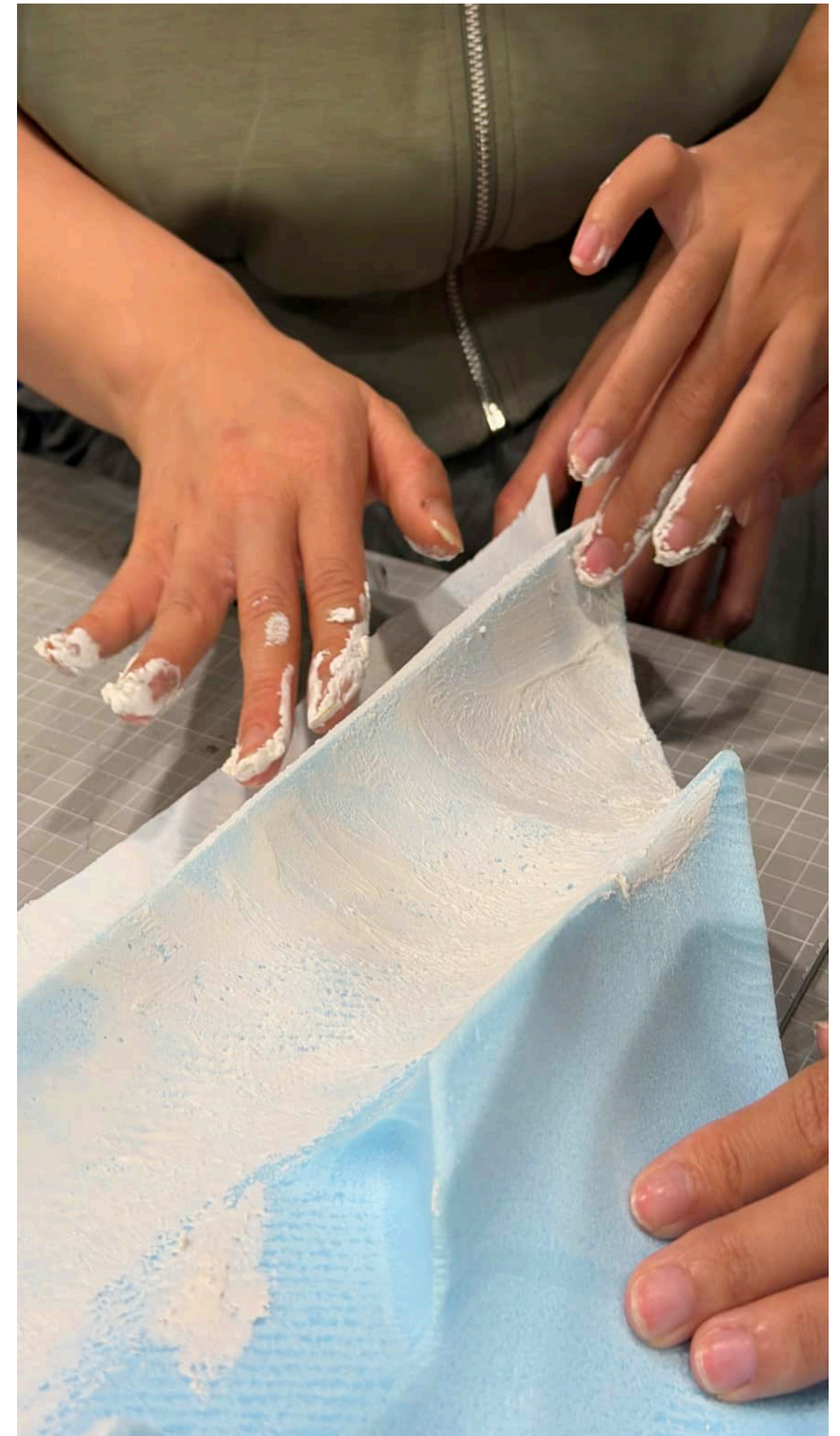
Project brief

ELECTIVE BASE RESEARCH ASSISTANT

The Elective Based Research Assistant (TFL) position was offered through the Tectonic Formation Lab research stream at RMIT University, coordinated by Marc Gibson. Running across Semester 1, 2026, as a 12-credit-point elective within the Master of Architecture program, the project centred on the research, development, and fabrication of a prototype architectural facade for Building 45, Lygon Street, Melbourne.

The facade system was conceived as a hybrid tectonic assembly combining cast Glass Fibre Reinforced Concrete (GFRC) panels, large-format polymer 3D-printed elements, and metal connectors — each material system contributing a distinct spatial and textural character to the overall composition.

My contribution spanned two intersecting workstreams: digital design development of complex ornamental surface geometries suitable for fabrication, and hands-on involvement in the physical fabrication pipeline — from CNC foam mould preparation through to casting, post-production, and final installation.



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FABRICATION PROCESS

Mould preparation | Post production | Installation

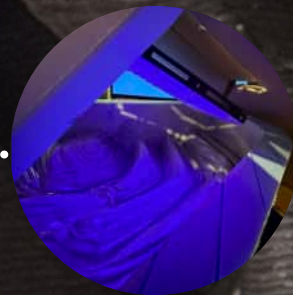
OVERALL PROCESS



CNC
foam milling



Mould
preparation



UV curing



Casting
process



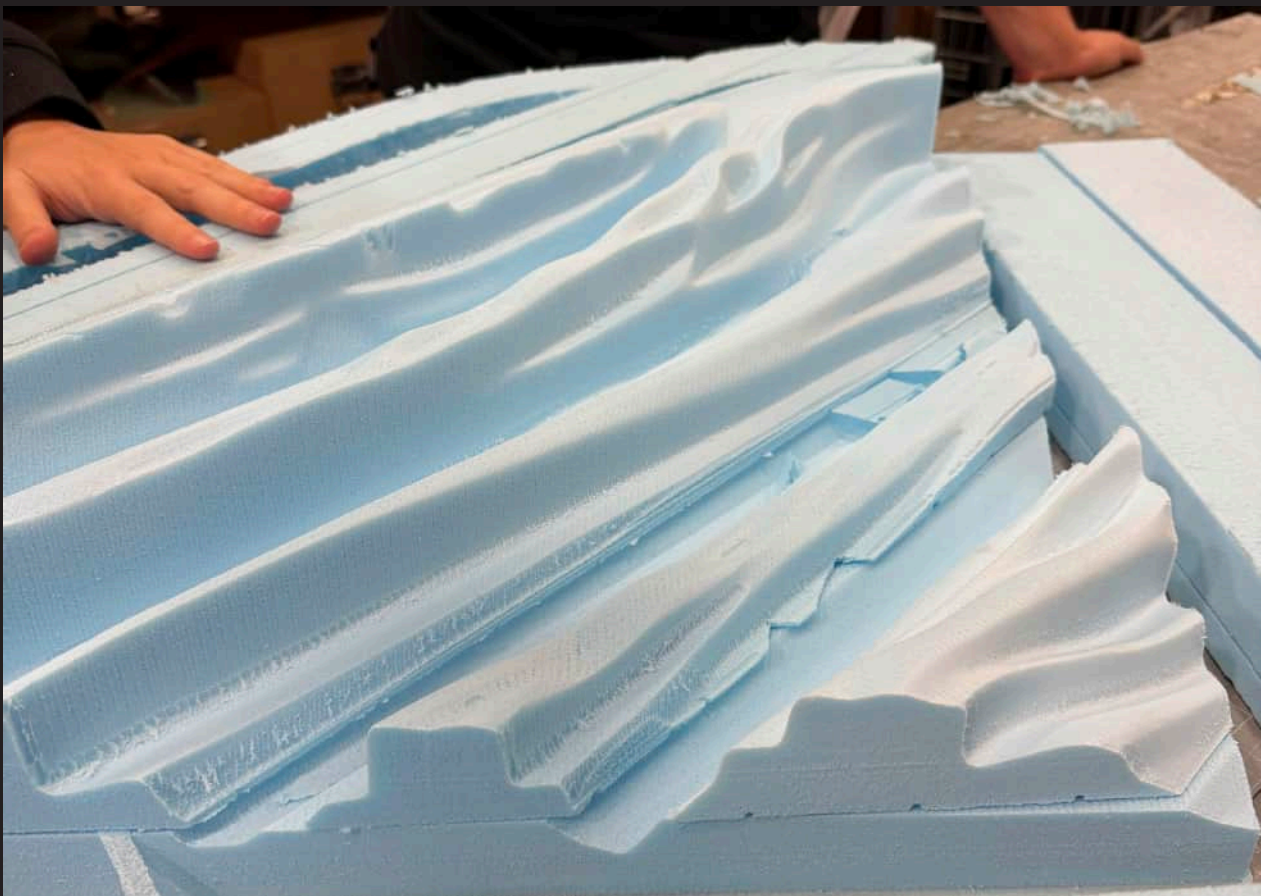
Post
production

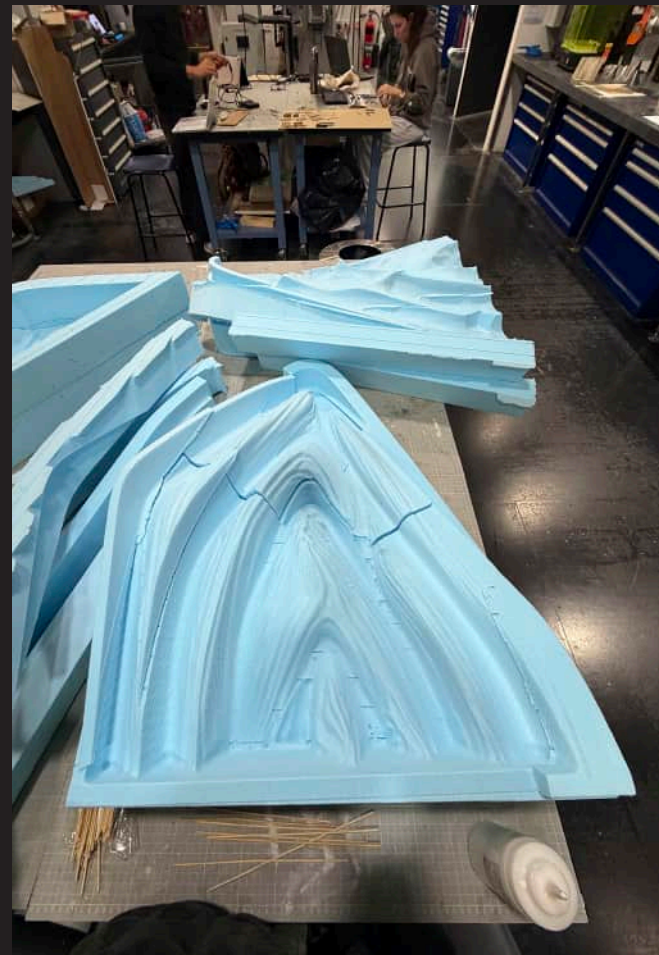


Final
assembly

CNC FOARM PARTS

The fabrication process began with CNC milling of large-format foam blocks to produce negative mould geometries. High-density blue insulation board was selected for its machinability, low weight, and dimensional stability under subsequent processing.

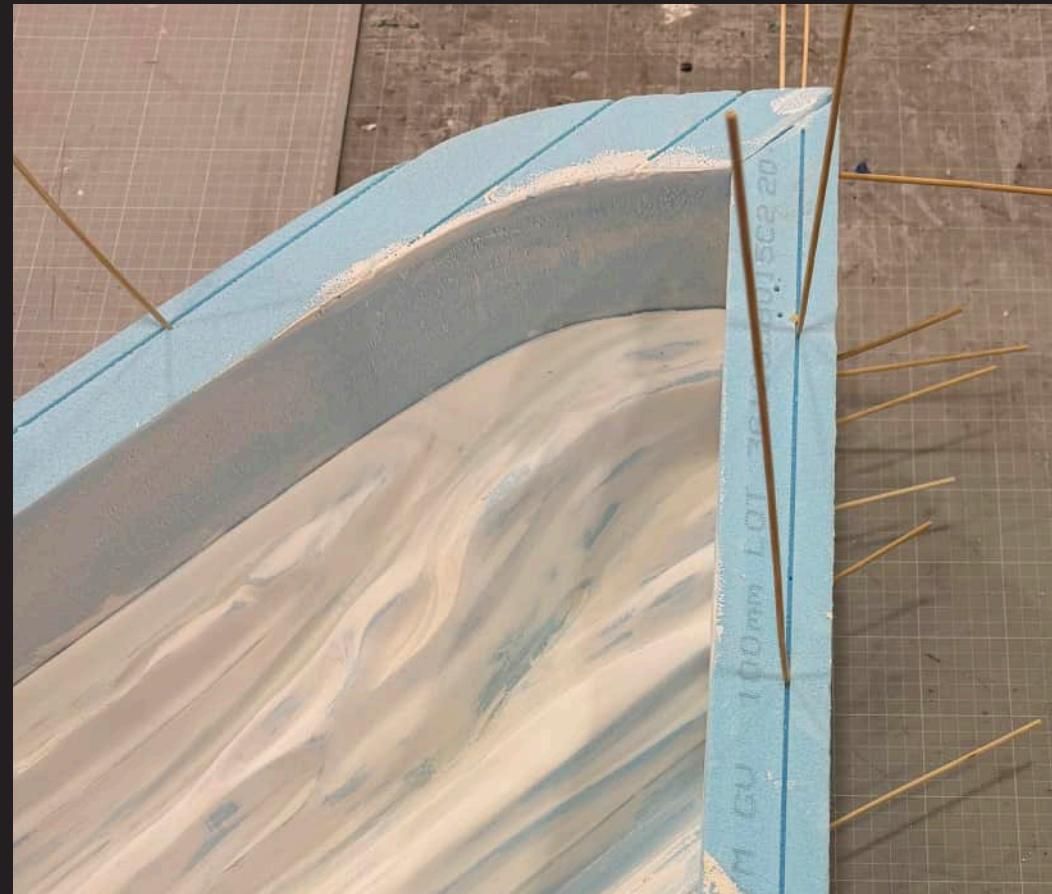




CNC FOARM PARTS

The CNC milling process translated complex digital surface geometry into physical foam components with high formal fidelity. Following milling, individual parts were manually refined — flash edges trimmed with a utility knife, and part-to-part alignment verified before proceeding to assembly.

The intricate contoured profiles milled into the foam surface directly captured the continuous folding and layered quality of the Aurora-generated surface geometry — a precise translation from digital mould negative to physical mould form.



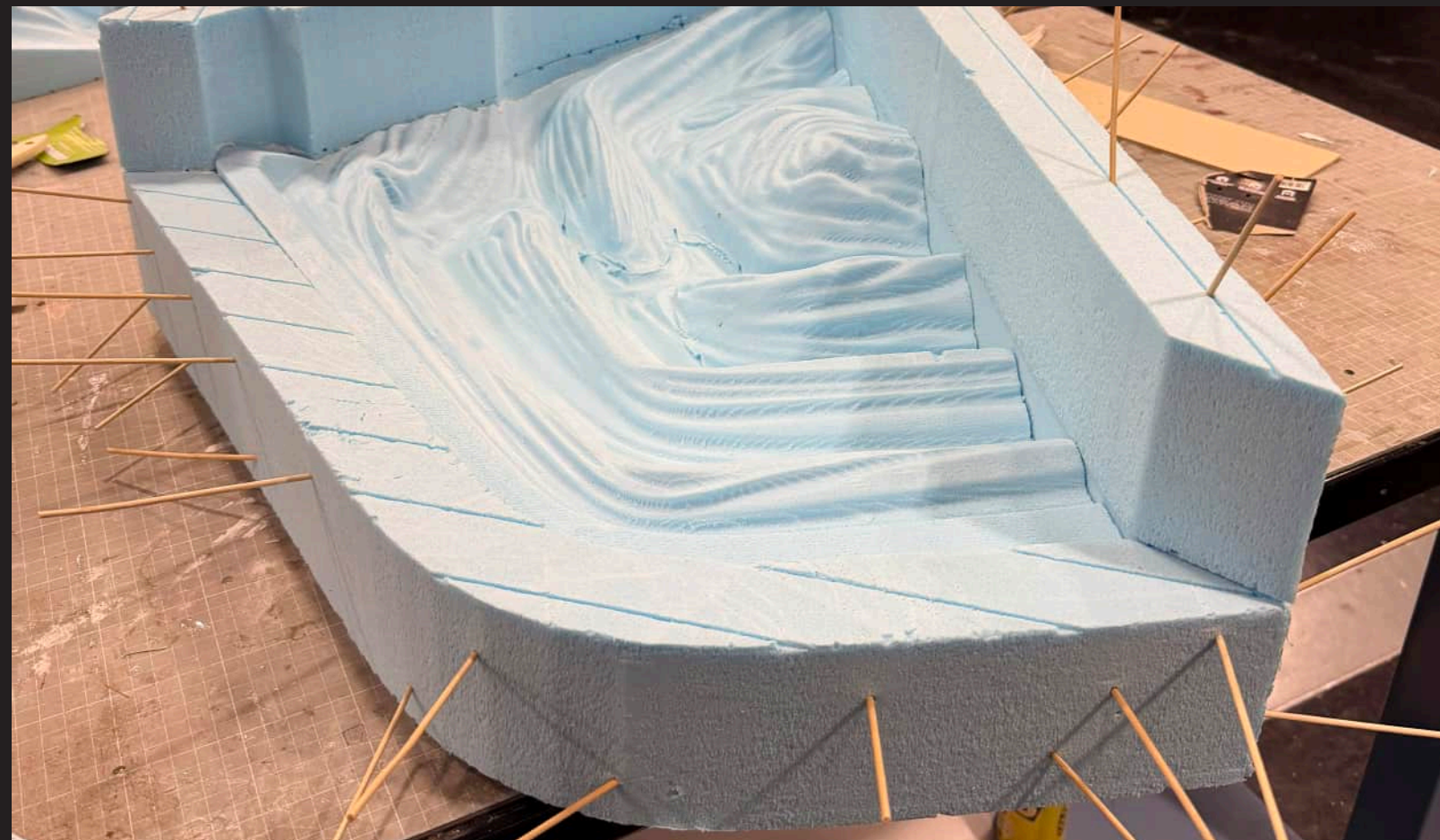
ASSEMBLY PARTS

PVA GLUE & BAMBOO STICKS

With individual CNC-milled foam components produced, the mould structure was assembled using PVA adhesive at part interfaces, with bamboo sticks inserted as mechanical reinforcement across joints — preventing relative movement during casting and maintaining dimensional accuracy across the assembled mould profile.

Temporary plastic bubble padding was positioned beneath fragile mould sections to prevent deflection or damage during handling and subsequent processing steps.

The assembled mould was then inspected for surface gaps and joint discontinuities, which were addressed in the following plaster patching phase.



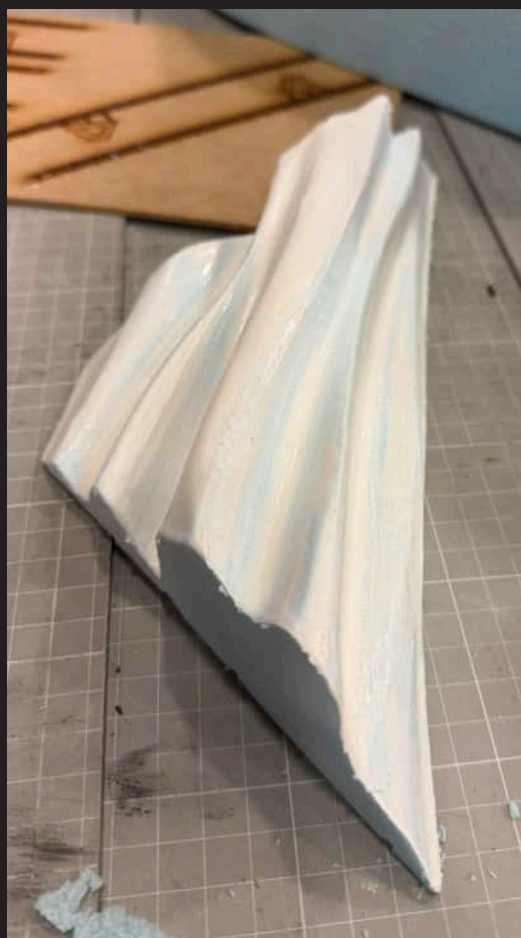


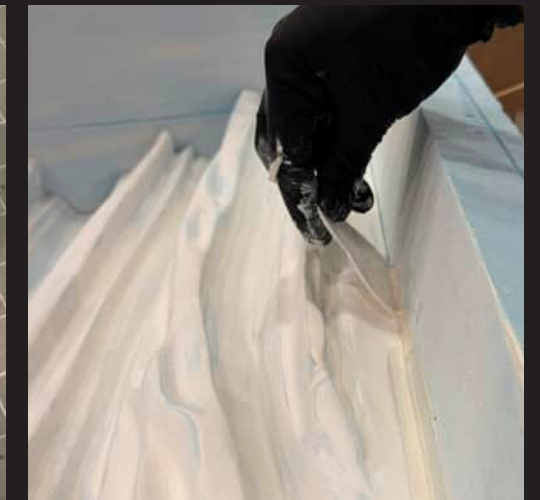
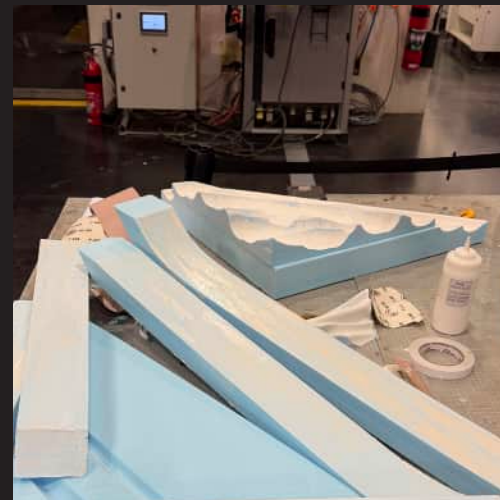
PLASTER PATCHING

Following mould assembly, all joint lines, surface voids, and CNC artefacts were filled using Uni-Pro Multi-Purpose Filler. This stage was critical in producing a continuous, seamless mould surface — any discontinuity left unresolved would transfer directly to the cast GFRC panel face.

Plaster was applied by hand and worked into the mould geometry using fingertips and spatulas, carefully following the surface contour. Once set, patched areas were sanded flush to the surrounding foam surface, eliminating any raised edges or surface steps.

This careful, iterative patching process ensured the mould surface was free of joints and defects before resin sealing.



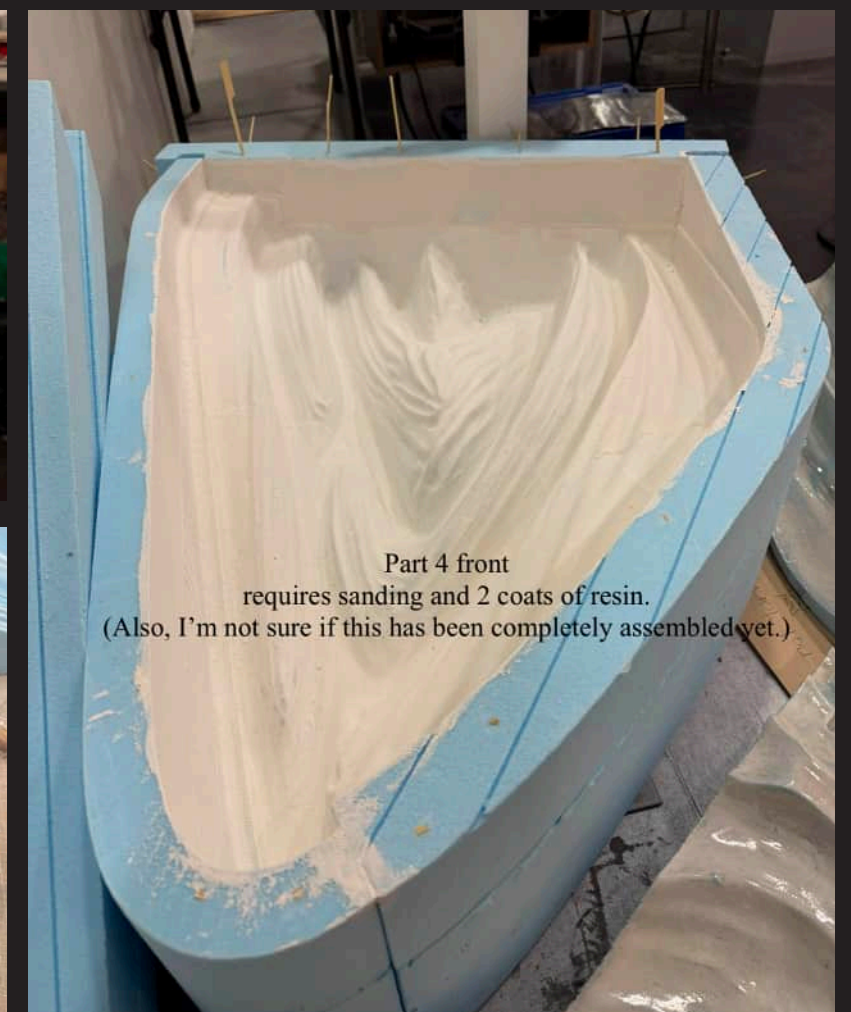


Water&Plaster combination

Smoothen surface

In areas where cutting and trimming had introduced surface irregularities, a water-diluted plaster mixture was applied to localised problem areas — allowing the material to self-level across the surface before being worked flat with a flexible blade.

Sanding across complex mould geometry required adaptive tool technique: employing angled, circular movements with the power sander, and selectively using the tool tip to address localised imperfections within recessed and curved areas. Where cuts left opaque or uneven surface residue, repeated localised repolishing cycles were performed to restore full surface uniformity.



Part 4 front
requires sanding and 2 coats of resin.
(Also, I'm not sure if this has been completely assembled yet.)

SANDING PLASTER

Plaster-patched mould surfaces were refined through a three-phase sanding sequence:

Rough Sanding (60-grit): Power sanding to reshape the surface geometry and remove major imperfections introduced during CNC milling or plaster application.

Edge Trimming: Manual trimming of residual flash edges using a clipping tool, followed by fine-grit sanding to smooth trimmed areas.

Fine Sanding (240-grit): Power sanding to produce the required surface smoothness, eliminating white discolouration and any remaining surface residue — preparing the mould for resin coating.

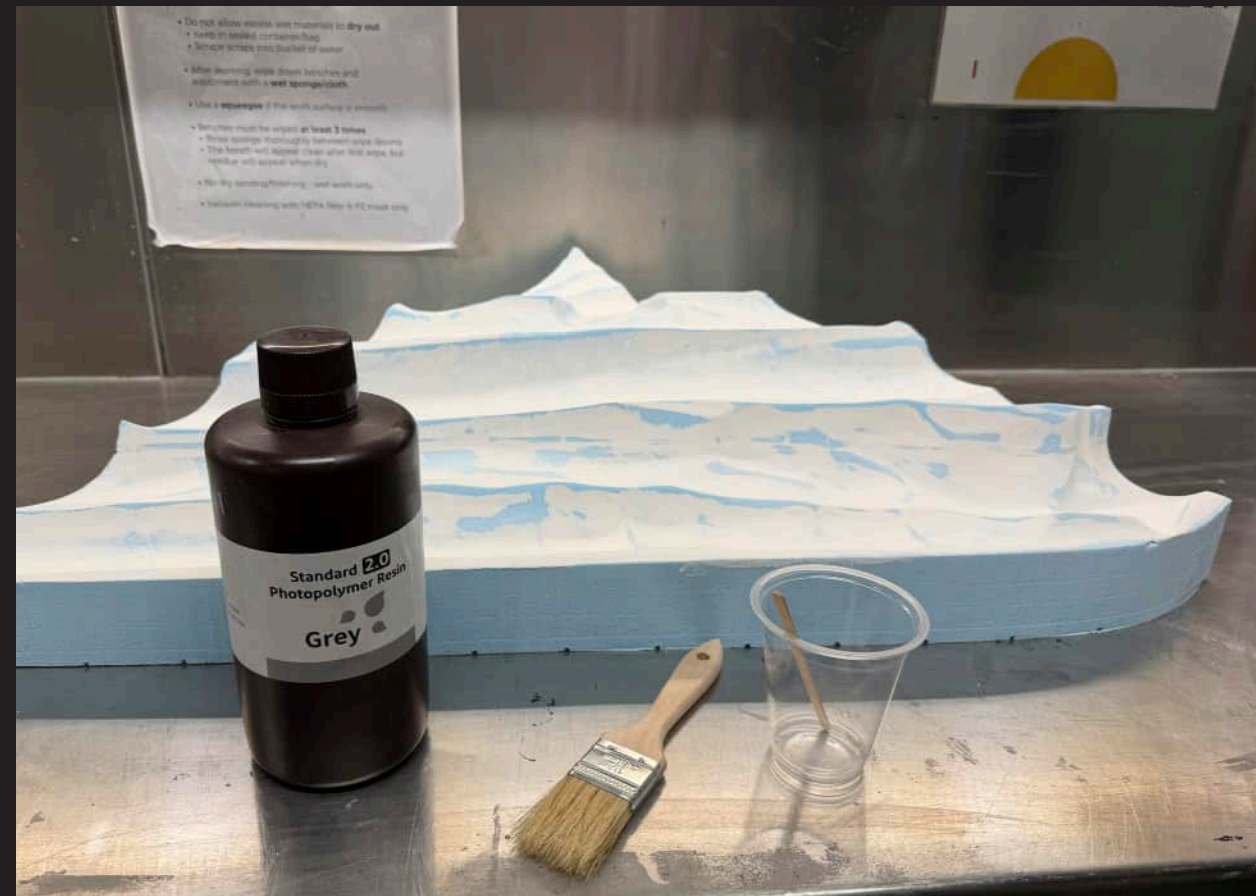
The resulting surface — smooth, continuous, and free of visible joints — provided the mould quality necessary to produce well-finished GFRC cast panels.



RASIN COATING

With the foam mould surface fully prepared, Standard Grey Photopolymer Resin was brush-applied across all internal mould surfaces. The resin coat served three functions: sealing the porous foam and plaster surface against moisture infiltration from the GFRC mix; providing a harder, more stable casting surface; and facilitating clean panel release during demoulding.

Resin was applied in controlled, even strokes using a natural bristle brush — working systematically across each mould section to ensure consistent coverage without pooling in surface recesses. Any uneven areas or resin runs were redistributed prior to curing.





CURING PROCESS

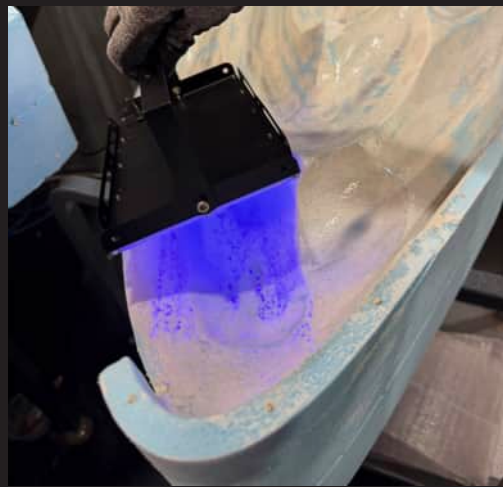
SUNLIGHT

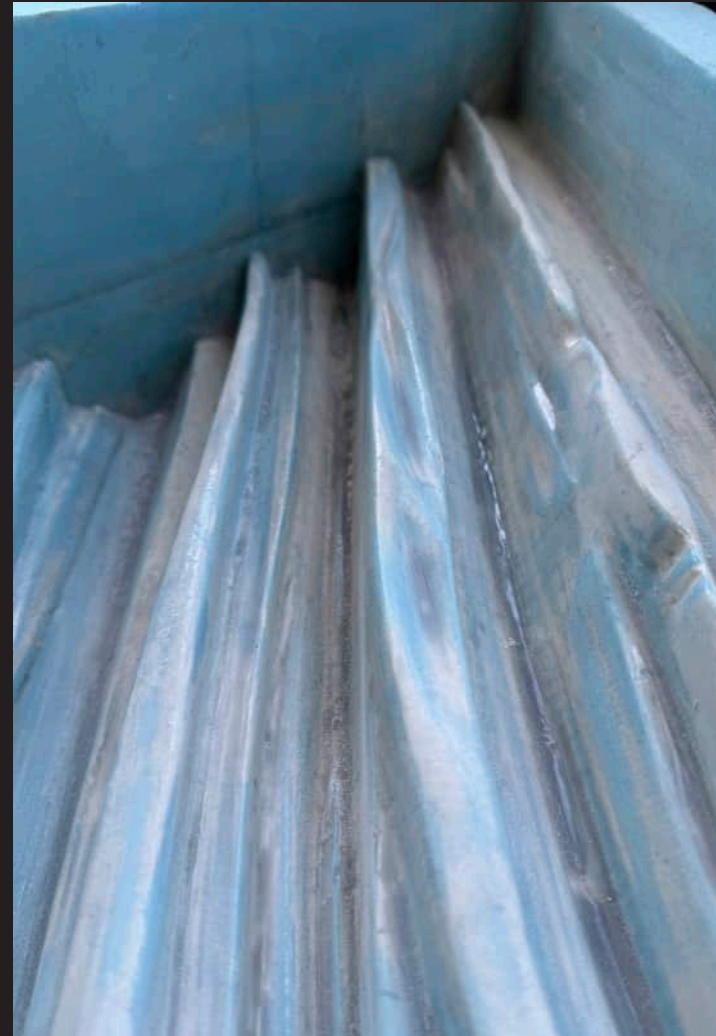
Resin-coated mould sections were transported outside and positioned in direct sunlight for initial cure. Moulds were oriented to maximise UV exposure, with periodic rotation to ensure even cure across all surfaces. Sunlight curing was deployed for full mould sections where direct lamp access was limited by scale.

UV LIGHT

Targeted UV lamp arrays were deployed for controlled, high-intensity curing of individual mould sections and areas requiring additional cure depth. Handheld UV lamps allowed precision application to recessed and complex surface areas unreachable by sunlight alone.

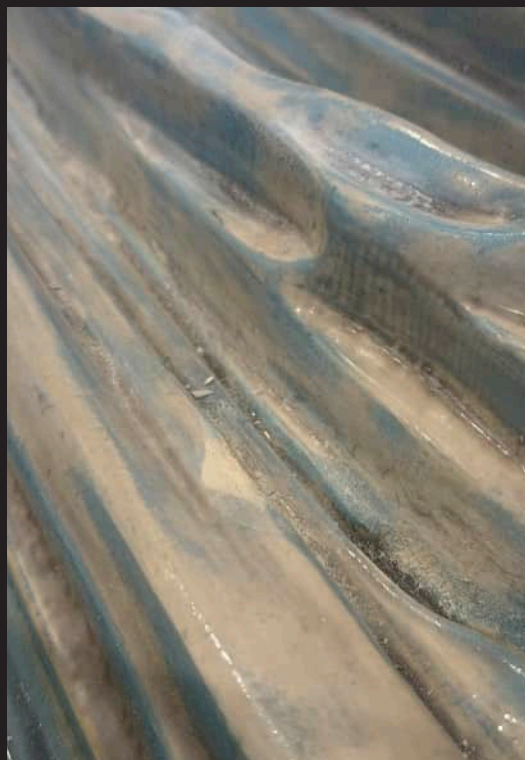
The combination of sunlight and UV lamp curing ensured complete through-cure of the resin layer, producing a hard, stable mould surface ready for the GFRC casting phase.





SANDING RESIN

Following full cure, resin-coated mould surfaces were sanded to refine texture and remove raised edges, resin drips, or inconsistencies from application. Power sanding with 60-grit paper addressed bulk imperfections, followed by 240-grit finishing to produce a smooth, uniform surface across the mould interior.



CRACK FIX

In areas where the resin cure or handling had introduced hairline cracks — particularly at foam joint locations or near sharp geometric transitions — targeted repair was performed. Cracks were filled with fresh resin and spot-cured using a handheld UV lamp, then sanded flush to restore surface continuity before proceeding to casting.

CASTING PHASE

With all mould sections fully prepared, sealed, and inspected, the GFRC casting phase commenced. Alpha Ete-Lease 800 mould release agent was applied to all internal mould surfaces to ensure clean panel separation at demoulding.

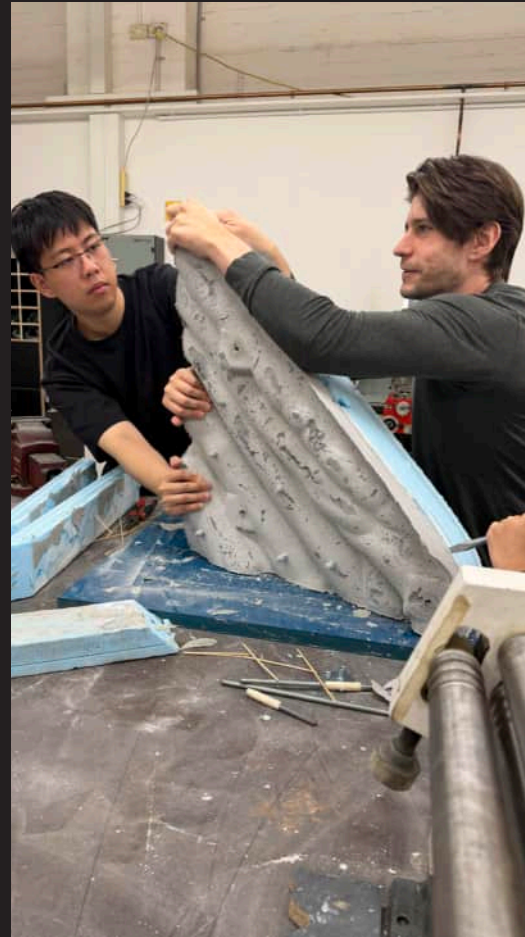
The GFRC mix — comprising Portland cement premix, water, iron oxide pigment, and chopped glass fibre strands — was prepared and hand-packed into the prepared moulds. The glass fibres, visible as a loose bundle prior to mixing, provided critical tensile reinforcement for the thin-shell panel geometry.

Casting required coordinated team work to ensure consistent mix application, adequate compaction into the complex surface recesses, and uniform panel thickness throughout. Test discs were cast simultaneously to evaluate mix consistency and surface quality across each batch. Marc Gibson supervised the mixing and casting stages, ensuring correct ratios and technique throughout.





DEMOULDING

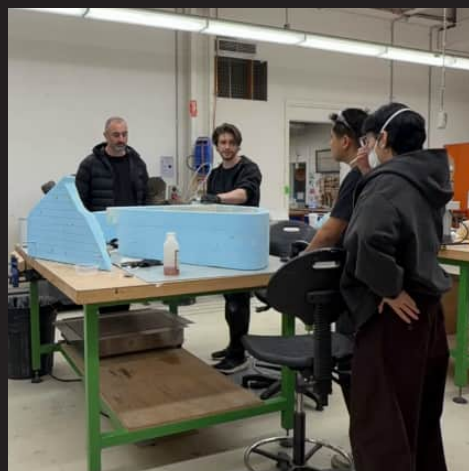


Following cure, cast GFRP panels were carefully separated from their foam mould negatives. Demoulding required progressive manual prying, controlled peeling, and lateral extraction — working from the mould edges inward to release each panel without fracturing the cast face.

The moment of demoulding revealed the direct material translation of the Aurora-generated digital surface into physical GFRP: the continuous folding, draping, and layered surface character of the computational model expressed in cast concrete. The aggregate texture and grey colouration of the GFRP surface produced an immediate material presence consistent with the monolithic language of the facade concept.

Close inspection of the demoulded panels confirmed the quality of the mould preparation — the flowing surface detail was faithfully reproduced, with only minor post-production refinement required.





POST PRODUCTION

Following demoulding, each cast GFRC panel was assessed and refined through a targeted post-production process. This stage addressed surface artefacts, residual release agent, minor cracking at thin sections, and areas requiring additional material fill.

Post-production tasks included: surface grinding to remove high spots and mould parting lines; crack repair using compatible patching mortar; brushing and cleaning to reveal the aggregate texture; and edge refinement to remove flash material.

This collaborative phase involved the full fabrication team, with individuals assigned to panels based on the nature and extent of post-production requirements. The goal was a consistent, installation-ready surface finish across all panels before final assembly.





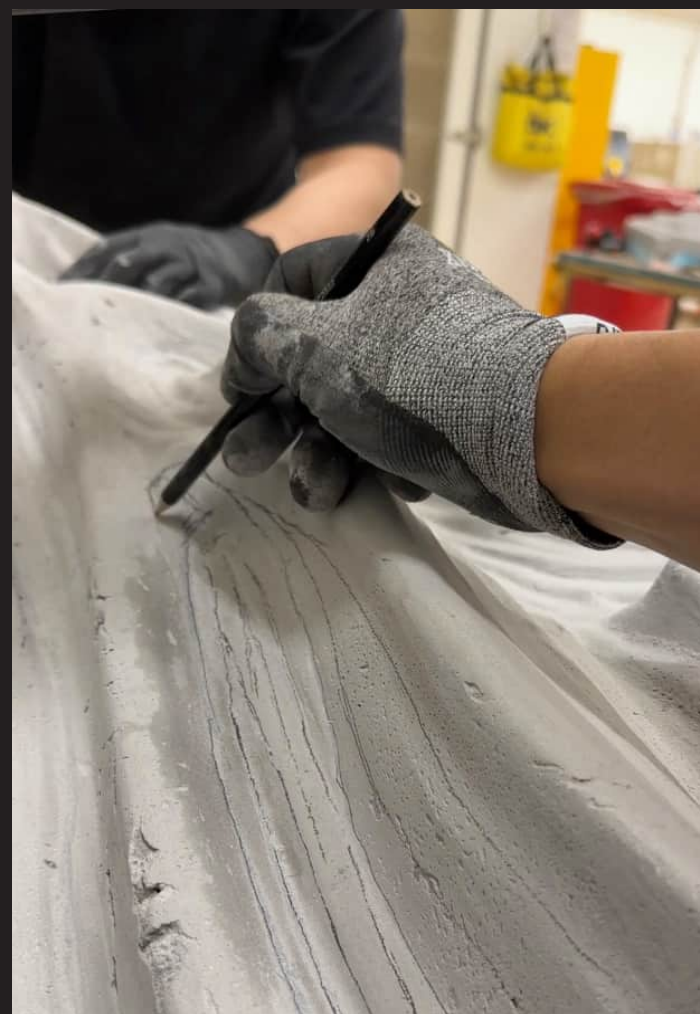
FINAL PROTOTYPE

The final prototype assembly brought together the two parallel fabrication workstreams: the cast GFRC panels carrying the monolithic surface language, and the large-format 3D-printed polymer structure providing the estranged structural counterpart.

Assembly required careful panel sequencing, structural alignment checks, and coordinated lifting of heavy GFRC components. Power tools were deployed to drill connection points, with panels positioned progressively from base to crown.

This stage marked the convergence of digital design development and physical fabrication — the first full-scale realisation of the TFL facade prototype

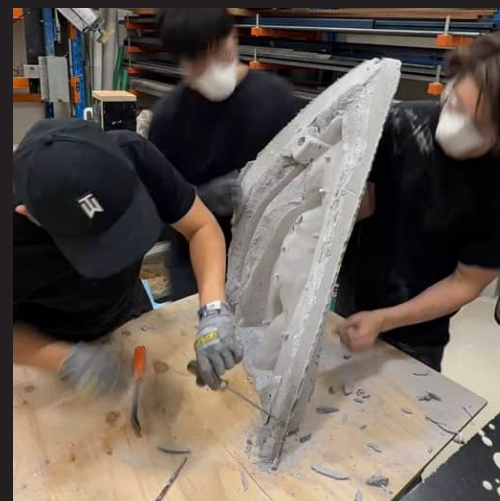




FINAL PROTOTYPE

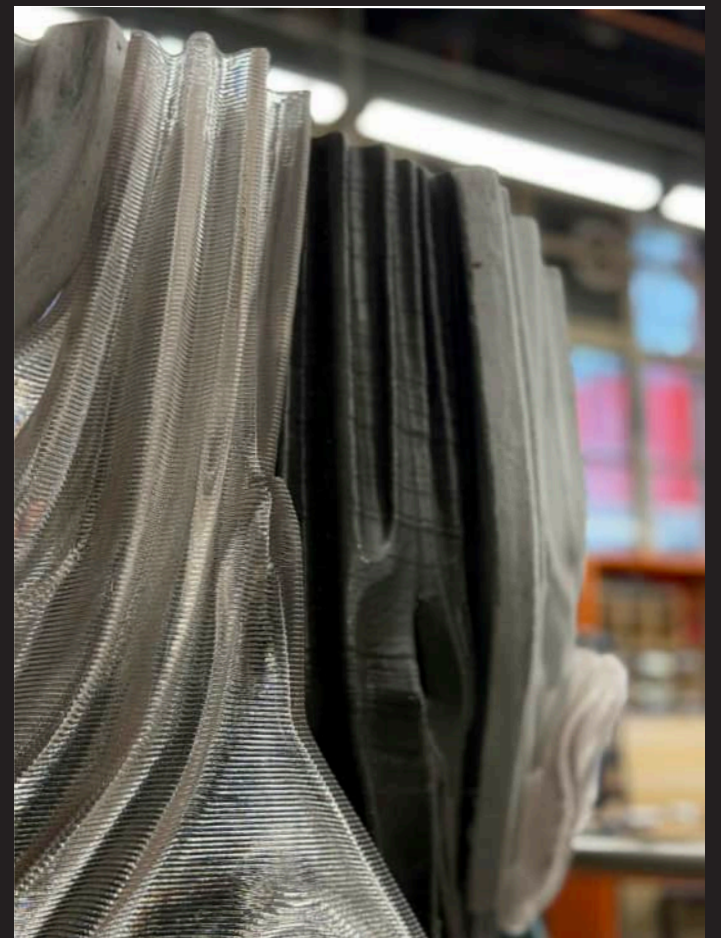
This stage marked the convergence of digital design development and physical fabrication — the first full-scale realisation of the TFL hybrid tectonic facade prototype

At this stage, the large-format 3D-printed polymer structure — produced by the parallel polymer printing team — was brought into direct juxtaposition with the GFRC panels. The visual contrast between the smooth, draped GFRC surface and the ribbed, directional polymer structure gave physical expression to the core tectonic dialogue of the project.





FINAL INSTALLATION





FINAL INSTALLATION

The final installation brought all fabricated components together at full assembly scale. Two key technical challenges emerged:

Unsmooth surface: Areas of joint mismatch at panel interfaces were addressed through in-situ patching and grinding to maintain surface continuity across the assembled facade.

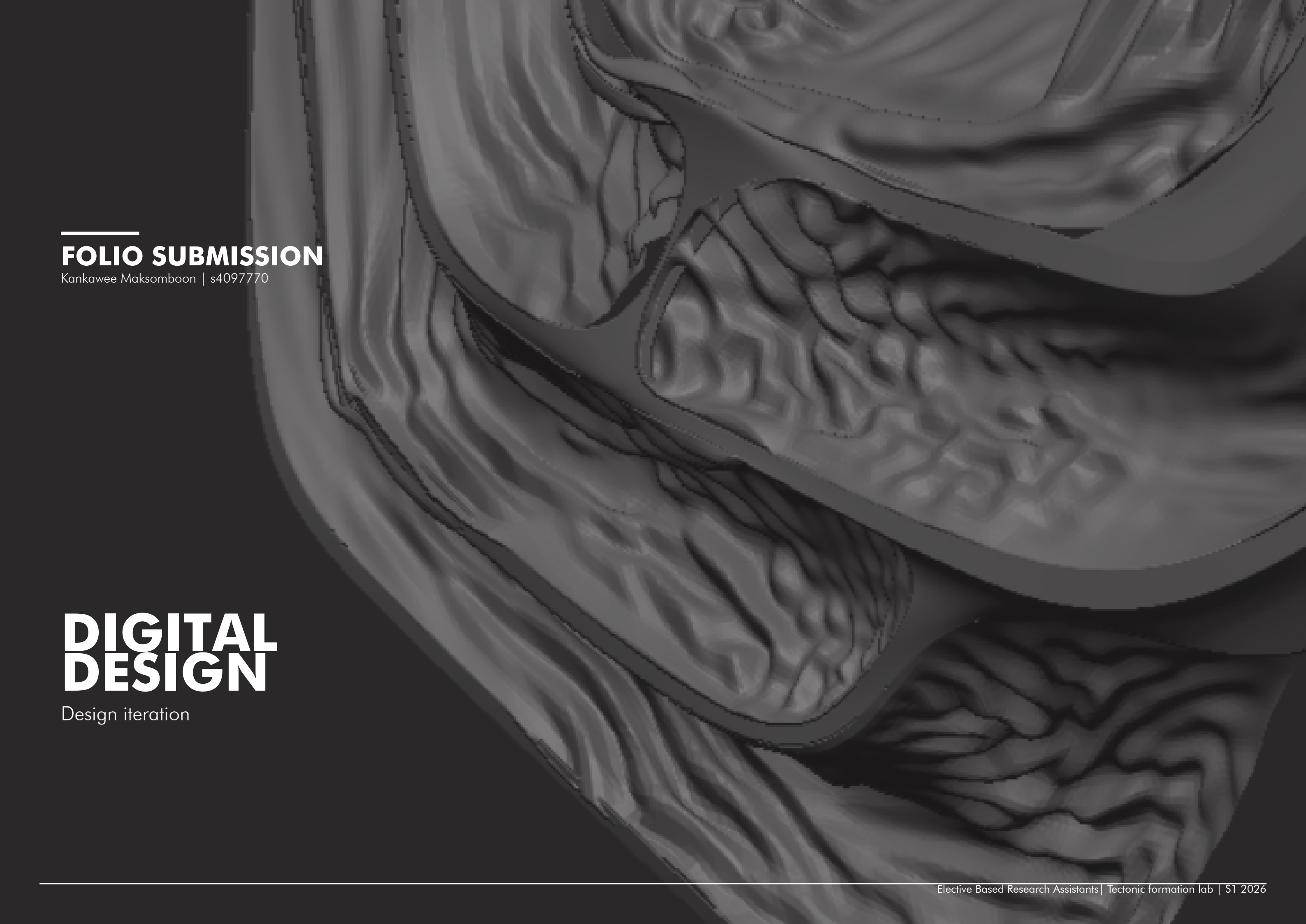
Joint

The structural connection between GFRC panels and the polymer frame was resolved through stainless steel bolt-and-bracket hardware. The joint zone — where the smooth monolithic GFRC surface meets the ribbed polymer structure — became a legible tectonic moment in the final assembly, expressing the hybrid character of the facade system.

Joint

The structural connection between GFRC panels and the polymer frame was resolved through stainless steel bolt-and-bracket hardware. The joint zone — where the smooth monolithic



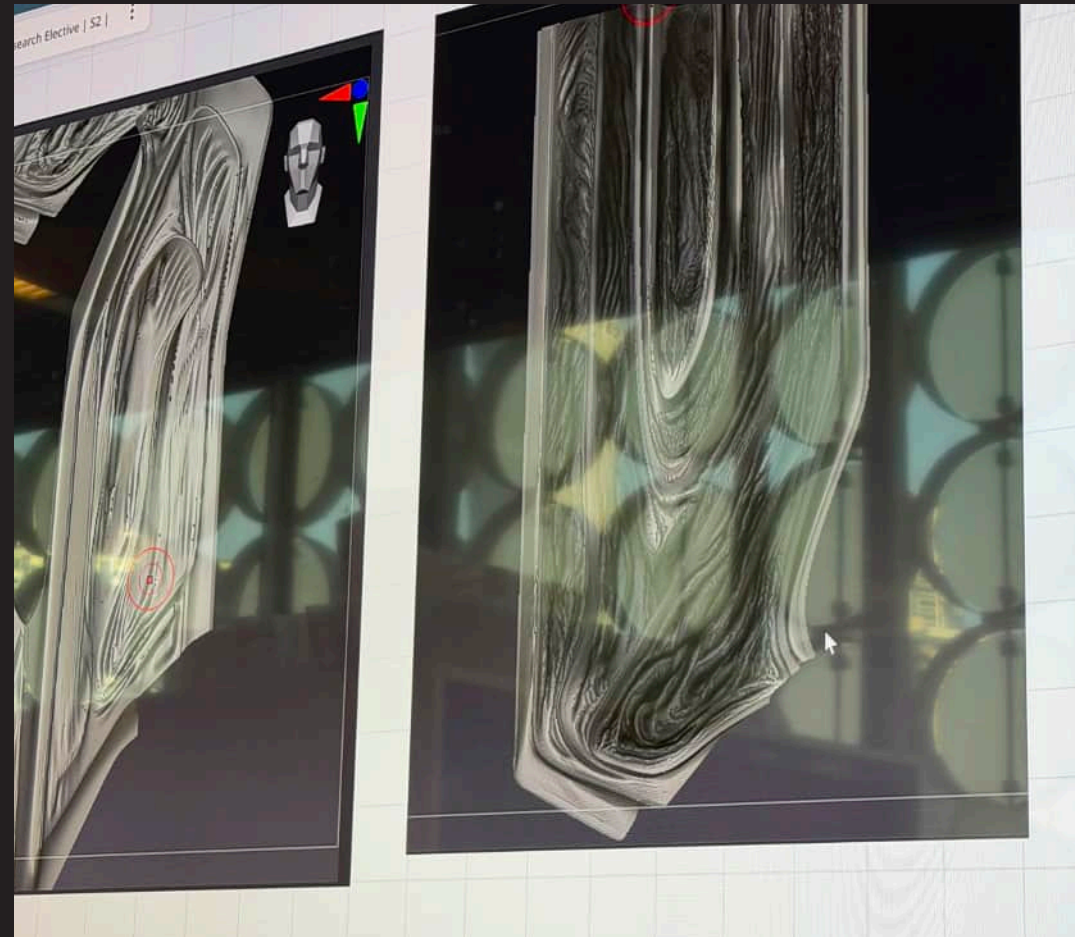
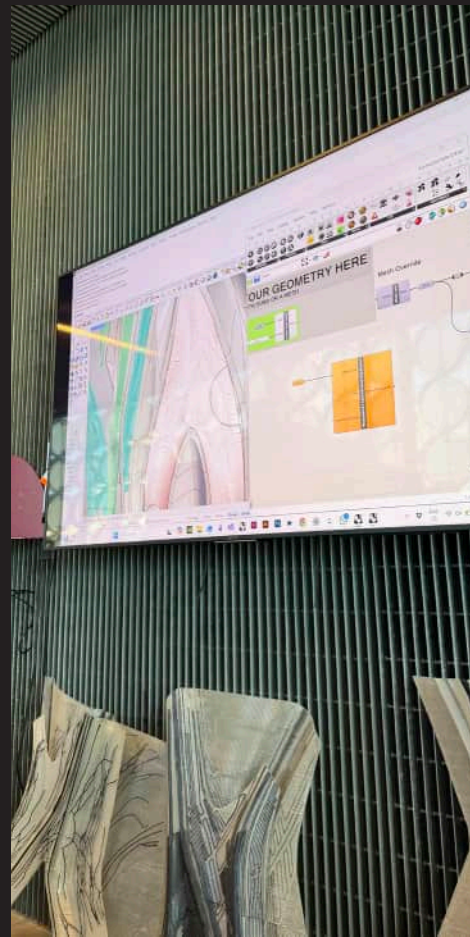


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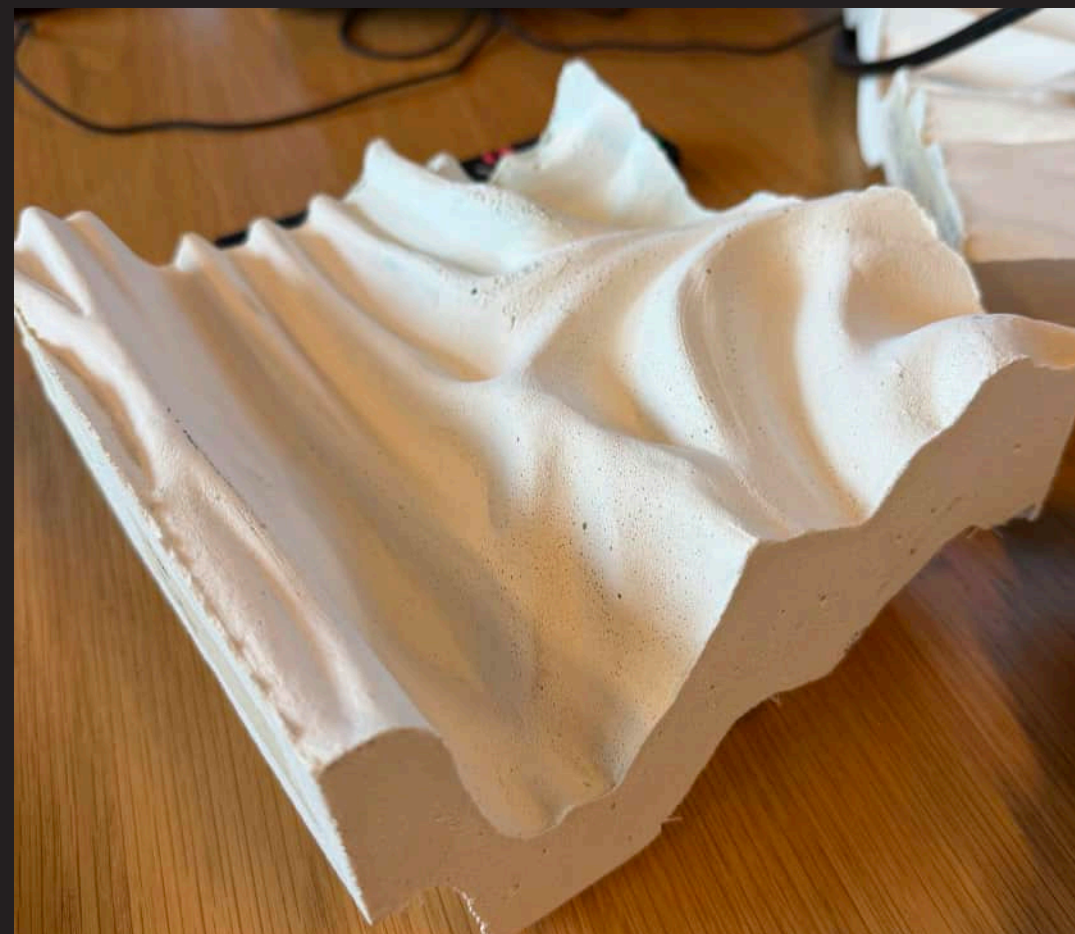
DIGITAL DESIGN

Design iteration

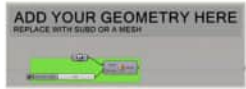


DIGITAL DESIGN BRIEF

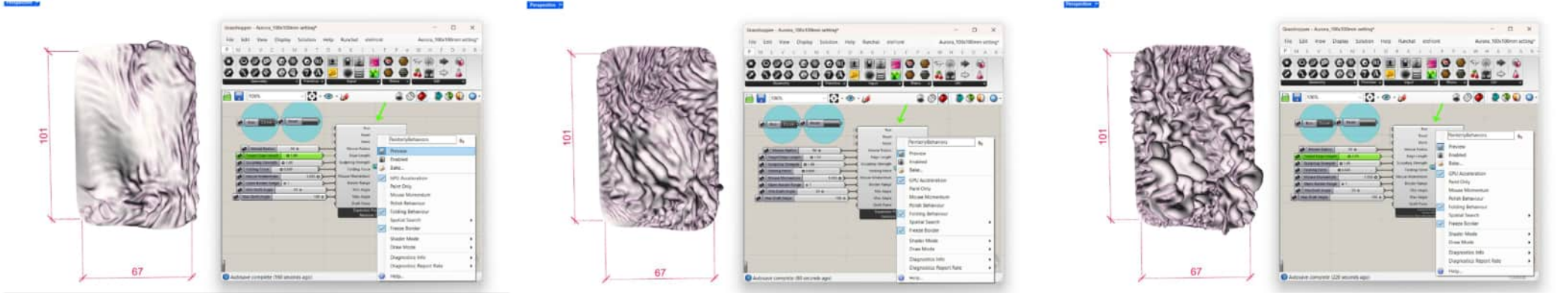
The brief required iterating surface geometry that was both visually resolved and practically achievable through the CNC foam mould fabrication process. This meant balancing surface complexity — fold depth, vein detail, directionality — against the fabrication constraints of mould milling and hand-casting.



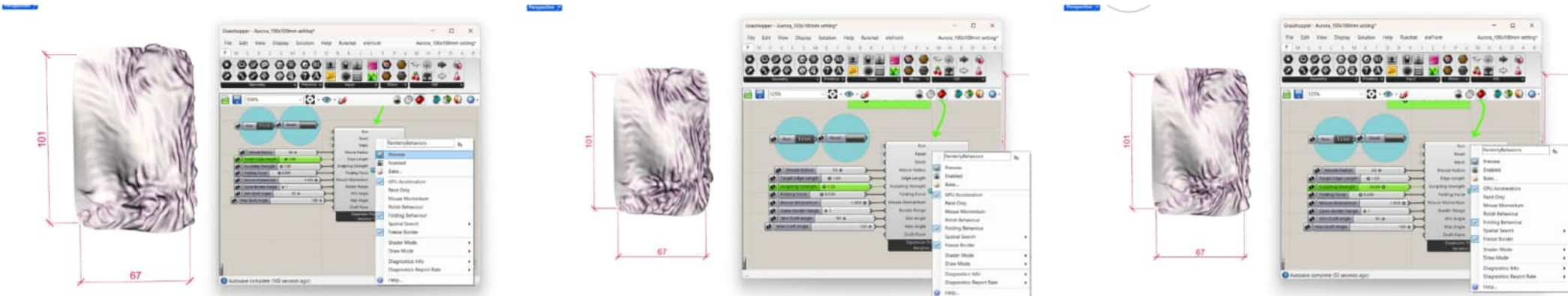
PARAMETER TESTING



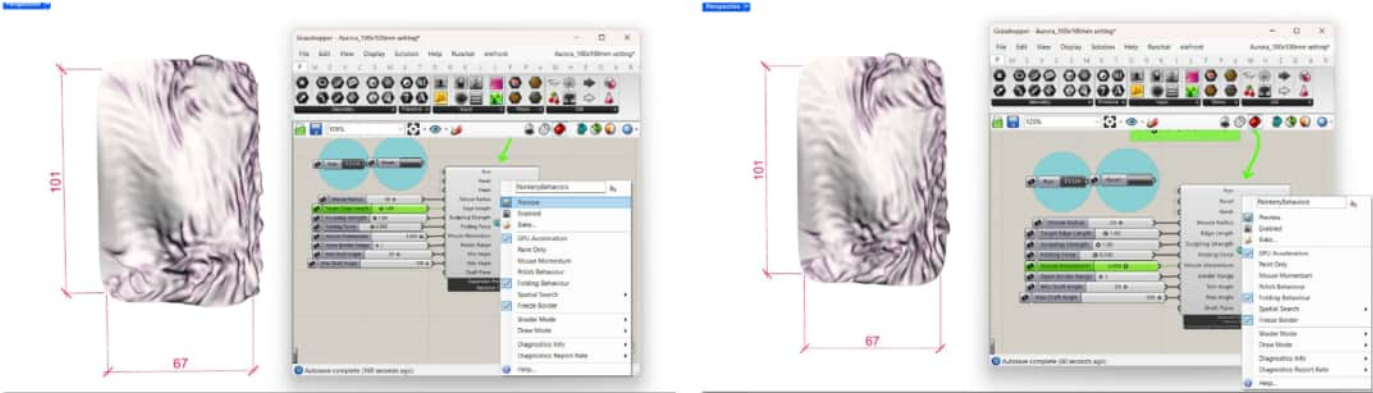
- Target Edge length



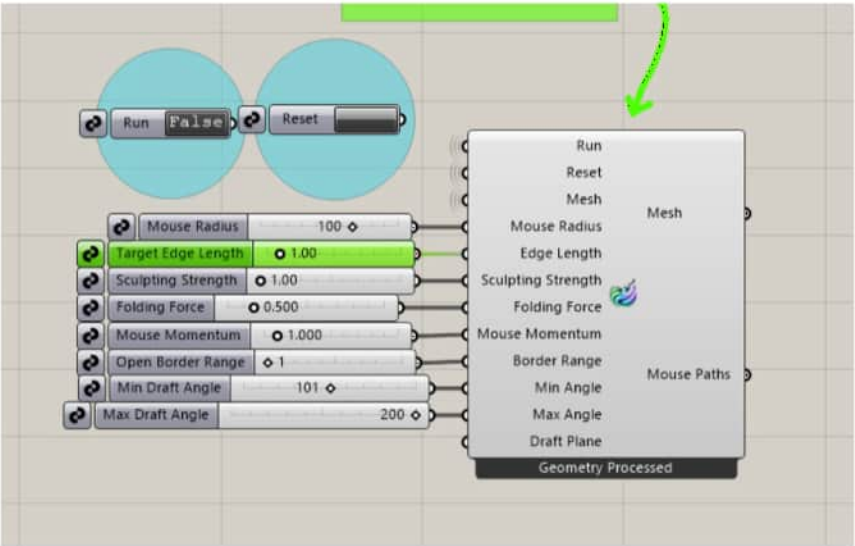
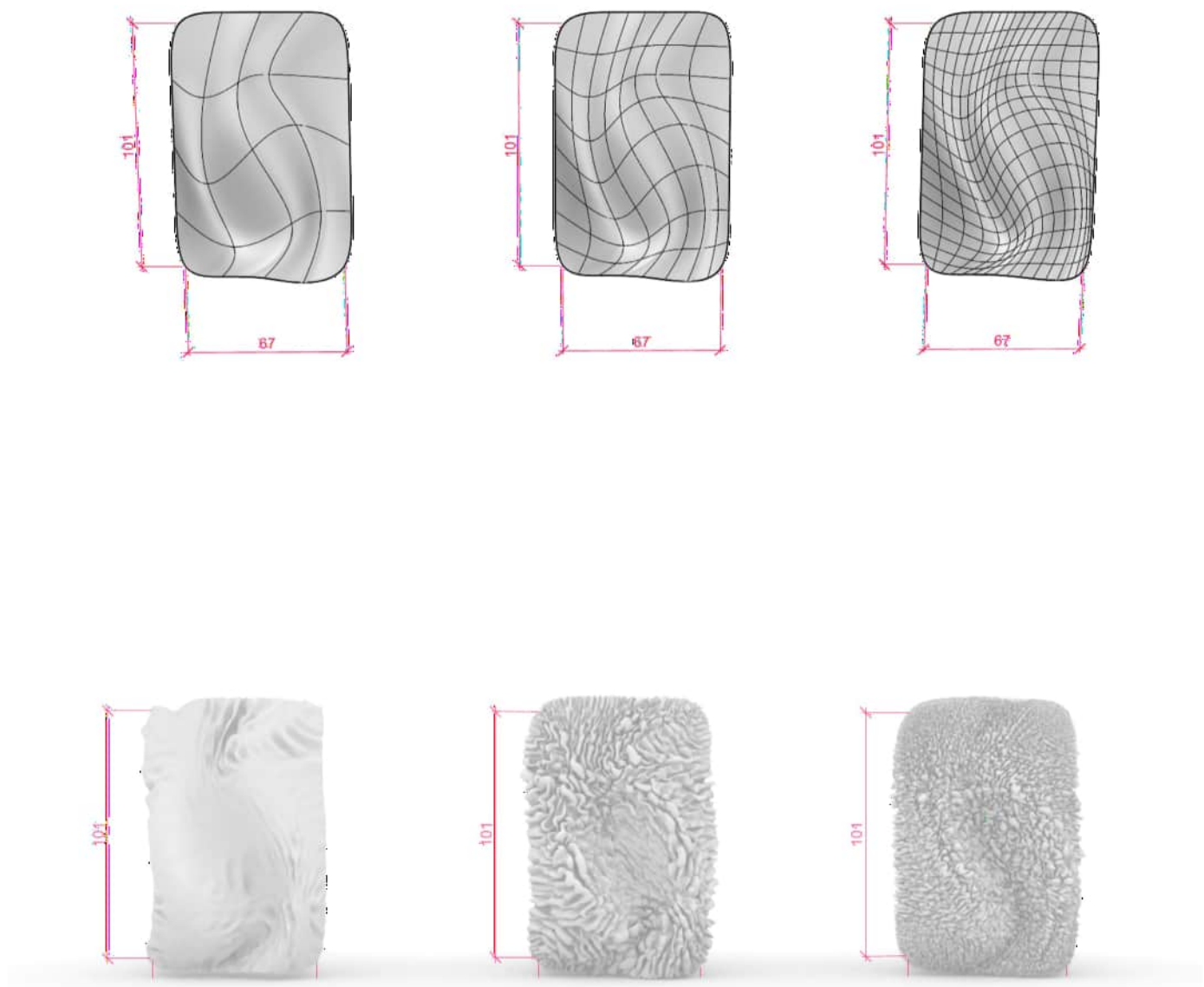
- Sculpture strength



- Sculpture strength

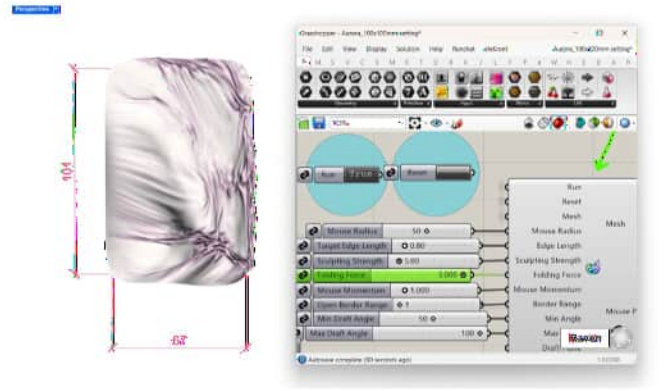
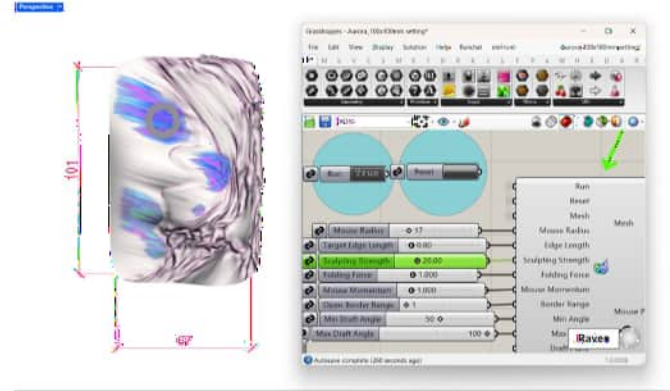
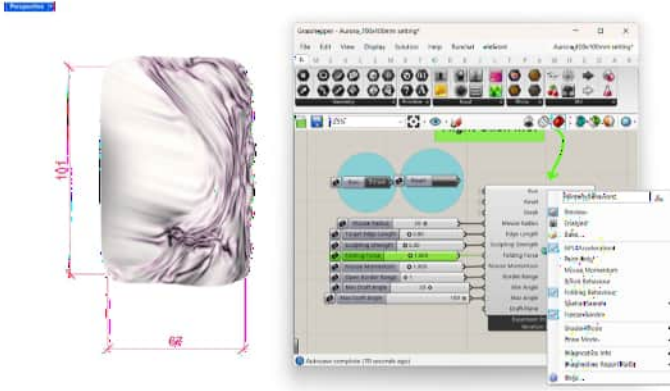
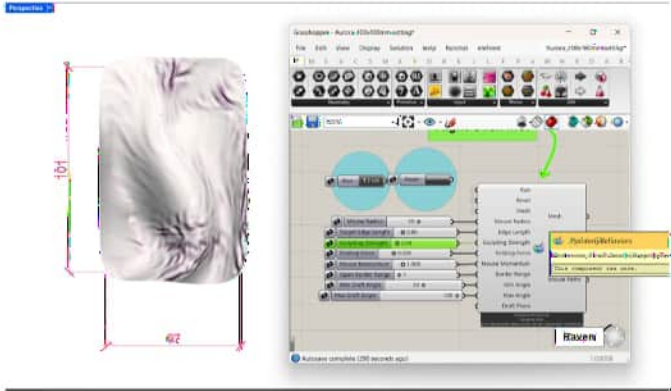


PARAMETER TESTING



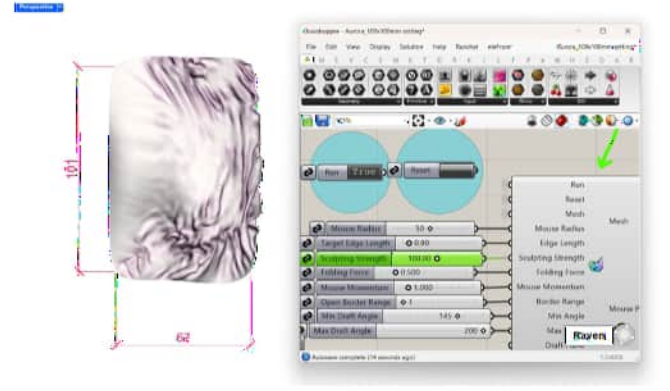
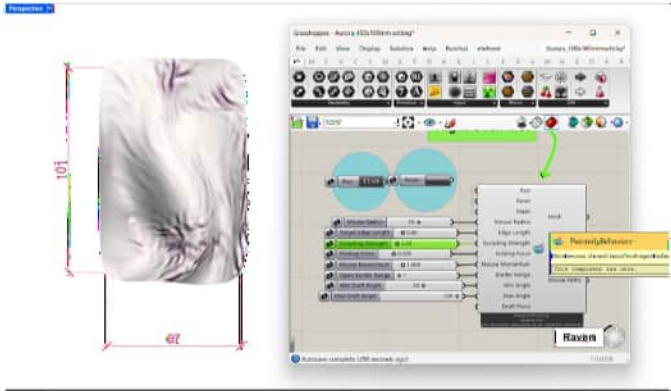
PARAMETER TESTING

- Lower Target Edge Length
- Increase Sculpting Strength
- Increase Folding Force
- Deal = 0.5g



Try to get small shape when fall

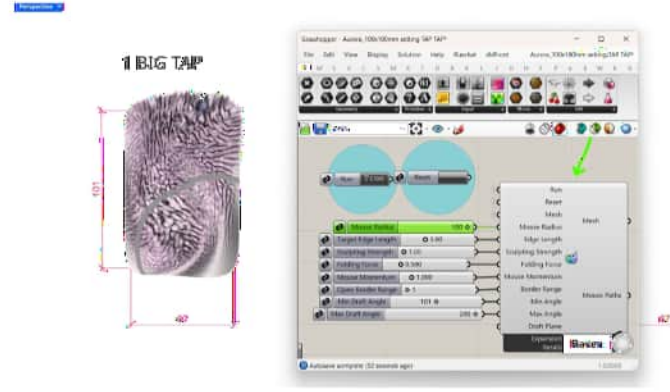
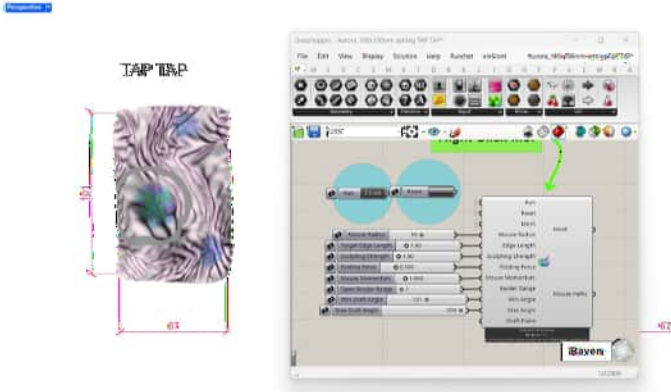
- Lower Target Edge Length
- Increase Sculpting Strength
- Increase Folding Force
- Deal = 0.5g



Detail work

Consistency small shape when

- Play with Target Edge Length + short click
- Lower number + Double click = Big when
- Higher number + Short click = Small when

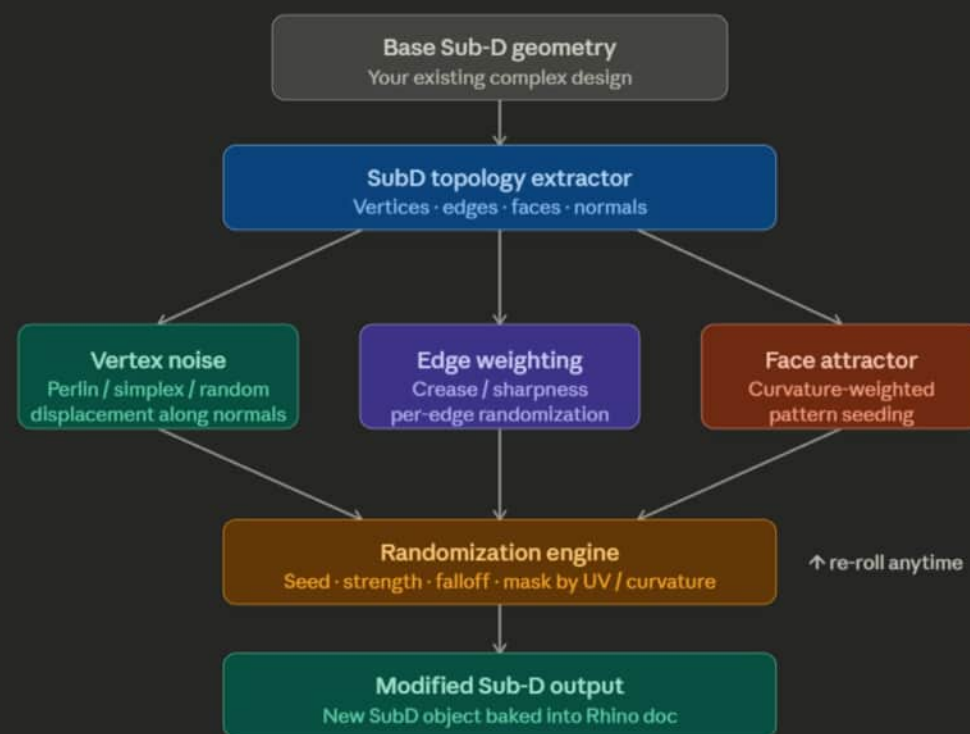


WHAT'S NEXT ??

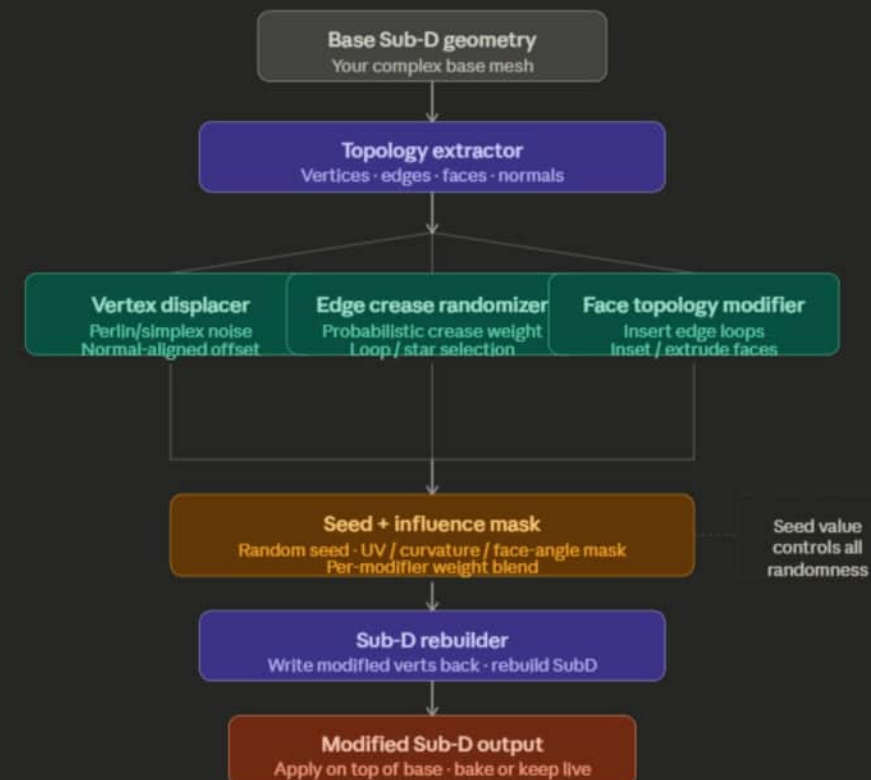
- Testing new features
- Create script Sub-d typology modifier:
- Prompt Adjust Sub-d or mesh surface provide configuration parameters including random Web for sub-d direction typology, and sub division.
- Prompt Adjust Sub-d or mesh surface that sensitive to multiple line attract I picked and provide configuration parameters including sub-d direction typology, and sub division.

Sub-D Surface Topology Modifier | Python script framework

FRAMEWORK_1



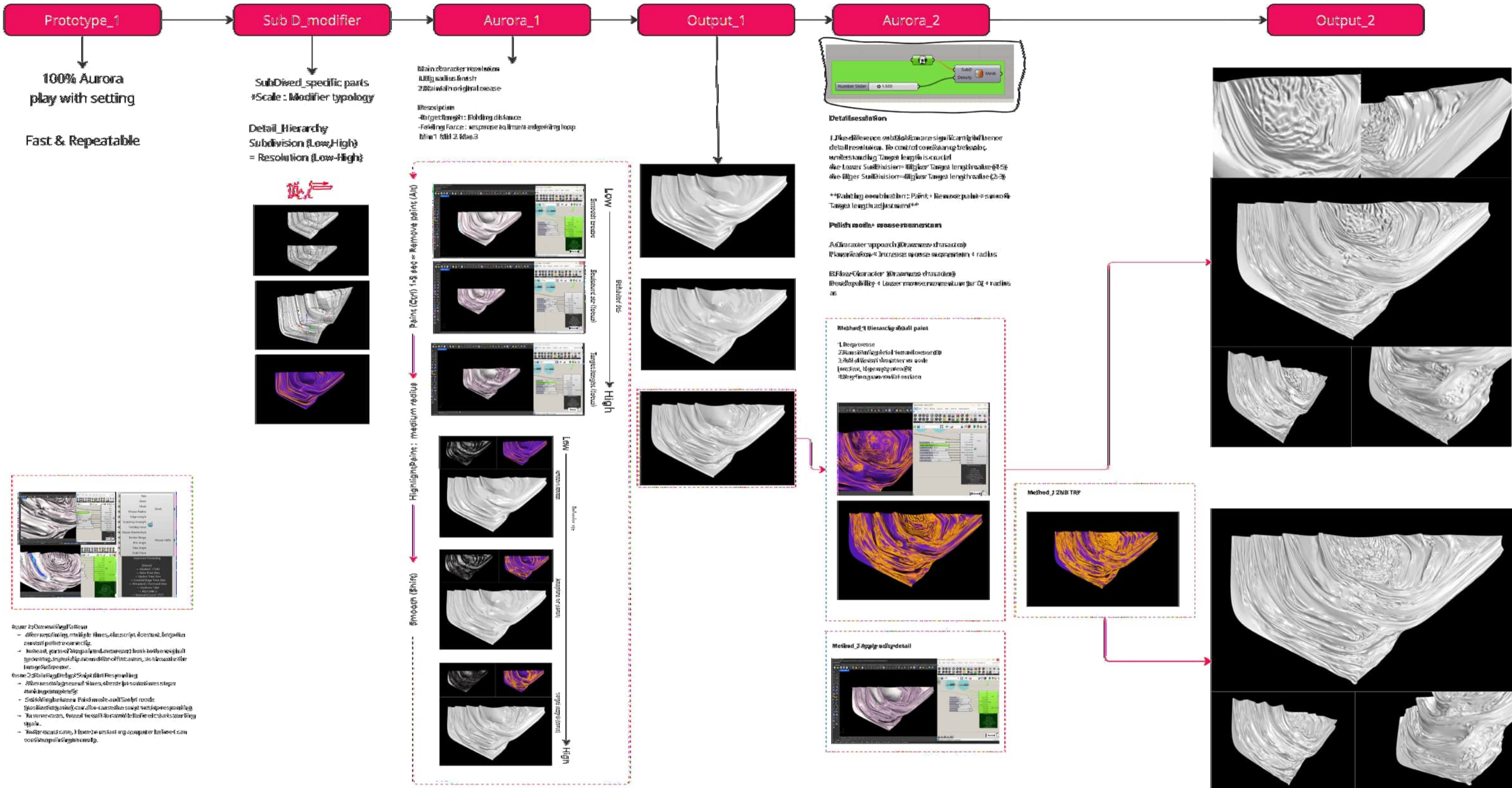
FRAMEWORK_2



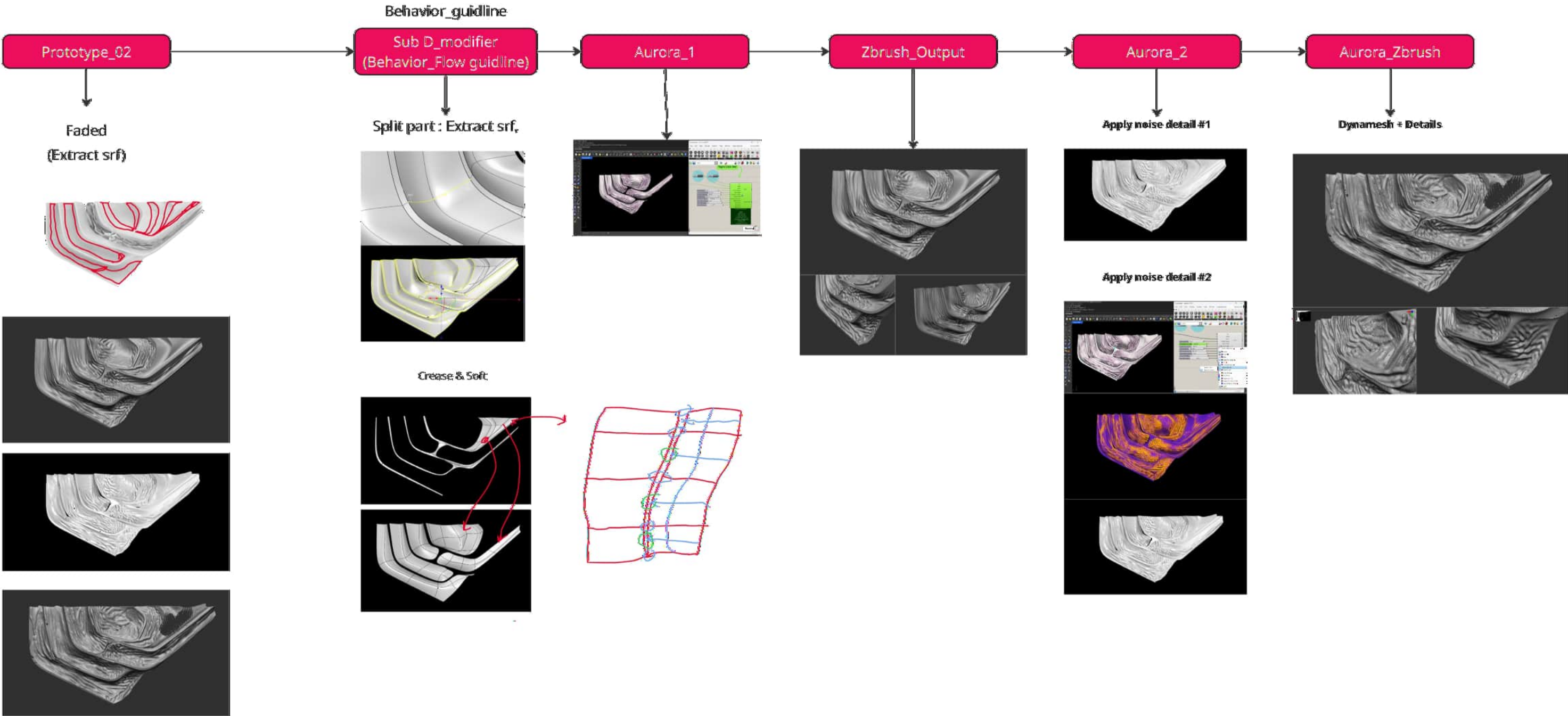
Python code

```
# Sub-D Topology Modifier for Rhino
# Base Sub-D geometry: Your complex base mesh
# Compatible with Rhino 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 653, 654, 655, 656, 657, 658, 659, 660, 661, 662, 663, 664, 665, 666, 667, 668, 669, 670, 671, 672, 673, 674, 675, 676, 677, 678, 679, 680, 681, 682, 683, 684, 685, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 718, 719, 720, 721, 722, 723, 724, 725, 726, 727, 728, 729, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793, 794, 795, 796, 797, 798, 799, 800, 801, 802, 803, 804, 805, 806, 807, 808, 809, 810, 811, 812, 813, 814, 815, 816, 817, 818, 819, 820, 821, 822, 823, 824, 825, 826, 827, 828, 829, 830, 831, 832, 833, 834, 835, 836, 837, 838, 839, 840, 841, 842, 843, 844, 845, 846, 847, 848, 849, 850, 851, 852, 853, 854, 855, 856, 857, 858, 859, 860, 861, 862, 863, 864, 865, 866, 867, 868, 869, 870, 871, 872, 873, 874, 875, 876, 877, 878, 879, 880, 881, 882, 883, 884, 885, 886, 887, 888, 889, 890, 891, 892, 893, 894, 895, 896, 897, 898, 899, 900, 901, 902, 903, 904, 905, 906, 907, 908, 909, 910, 911, 912, 913, 914, 915, 916, 917, 918, 919, 920, 921, 922, 923, 924, 925, 926, 927, 928, 929, 930, 931, 932, 933, 934, 935, 936, 937, 938, 939, 940, 941, 942, 943, 944, 945, 946, 947, 948, 949, 950, 951, 952, 953, 954, 955, 956, 957, 958, 959, 960, 961, 962, 963, 964, 965, 966, 967, 968, 969, 970, 971, 972, 973, 974, 975, 976, 977, 978, 979, 980, 981, 982, 983, 984, 985, 986, 987, 988, 989, 990, 991, 992, 993, 994, 995, 996, 997, 998, 999, 1000
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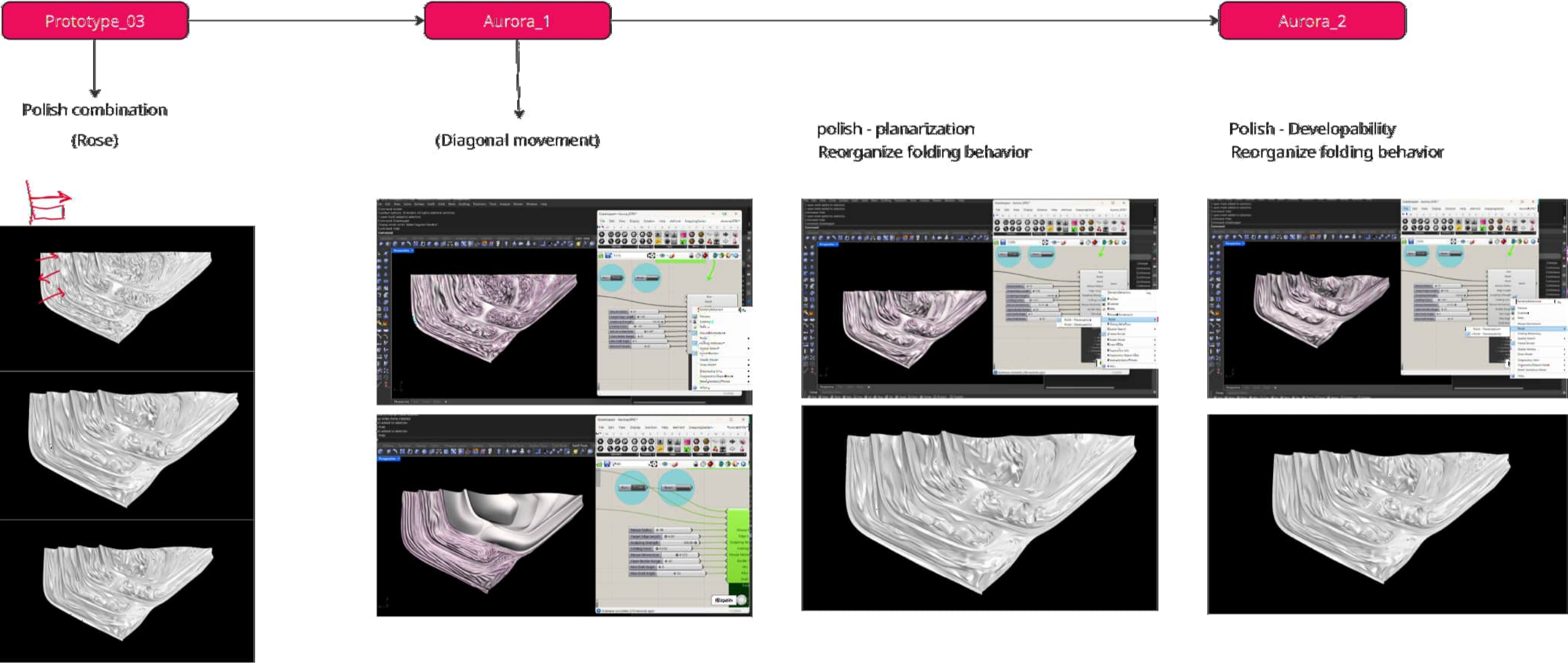

Design iteration_Prototype_1



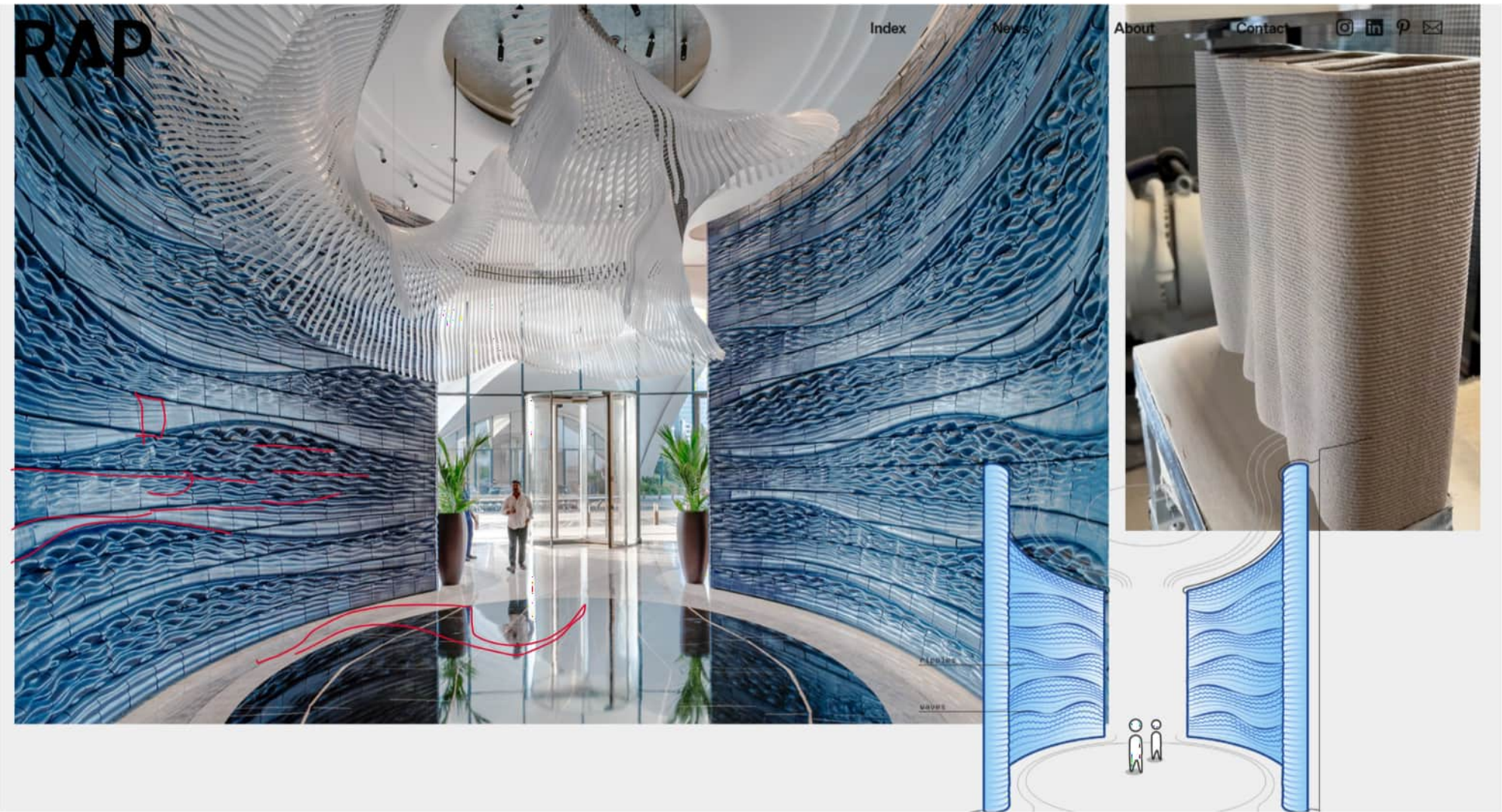
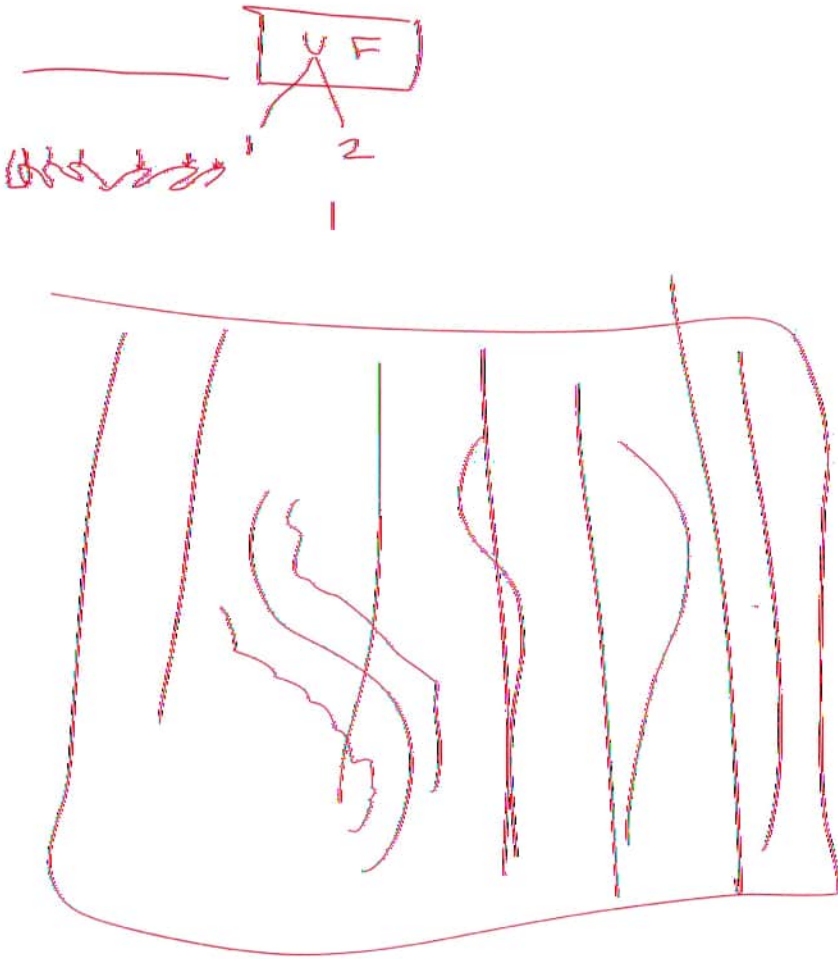
Design iteration_Prototype_2



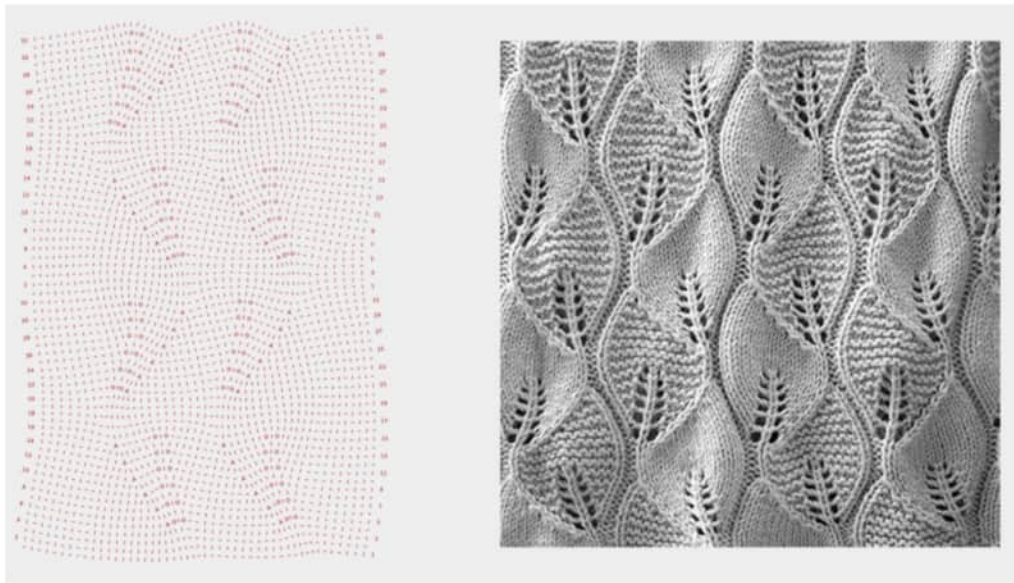
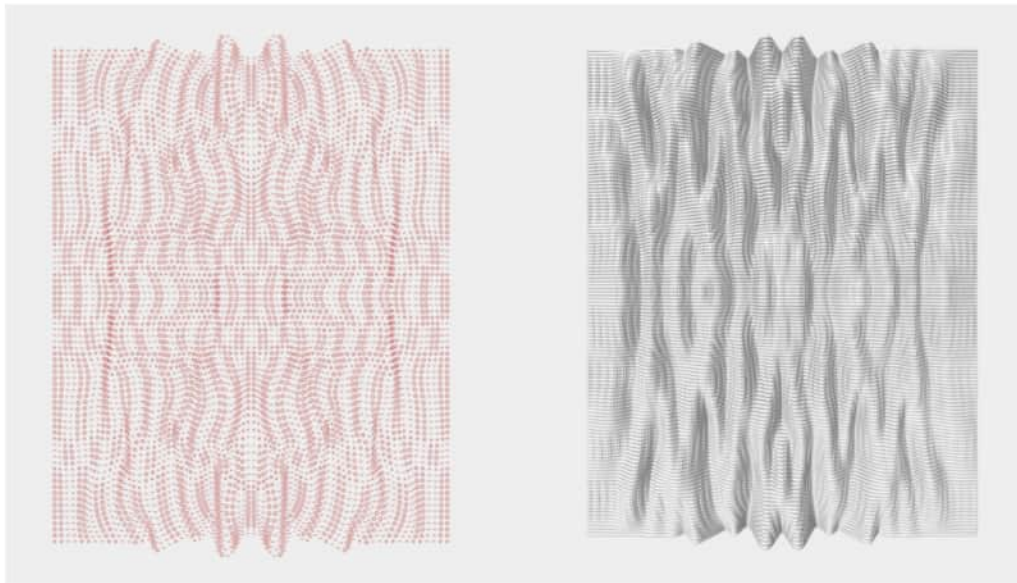
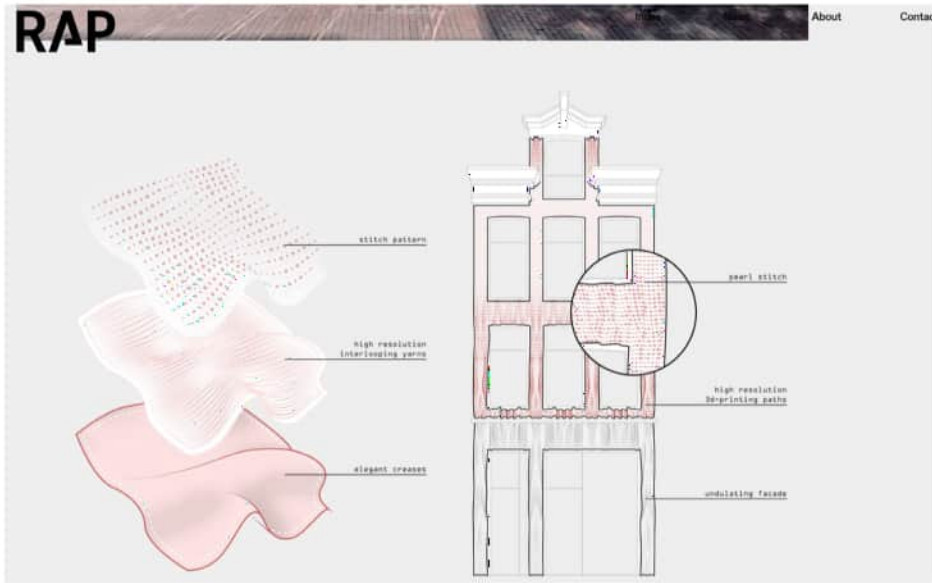
Design iteration_Prototype_3



Case study_ Studio RAP _ Ceramic facade



Case study_ Studio RAP _ Ceramic facade



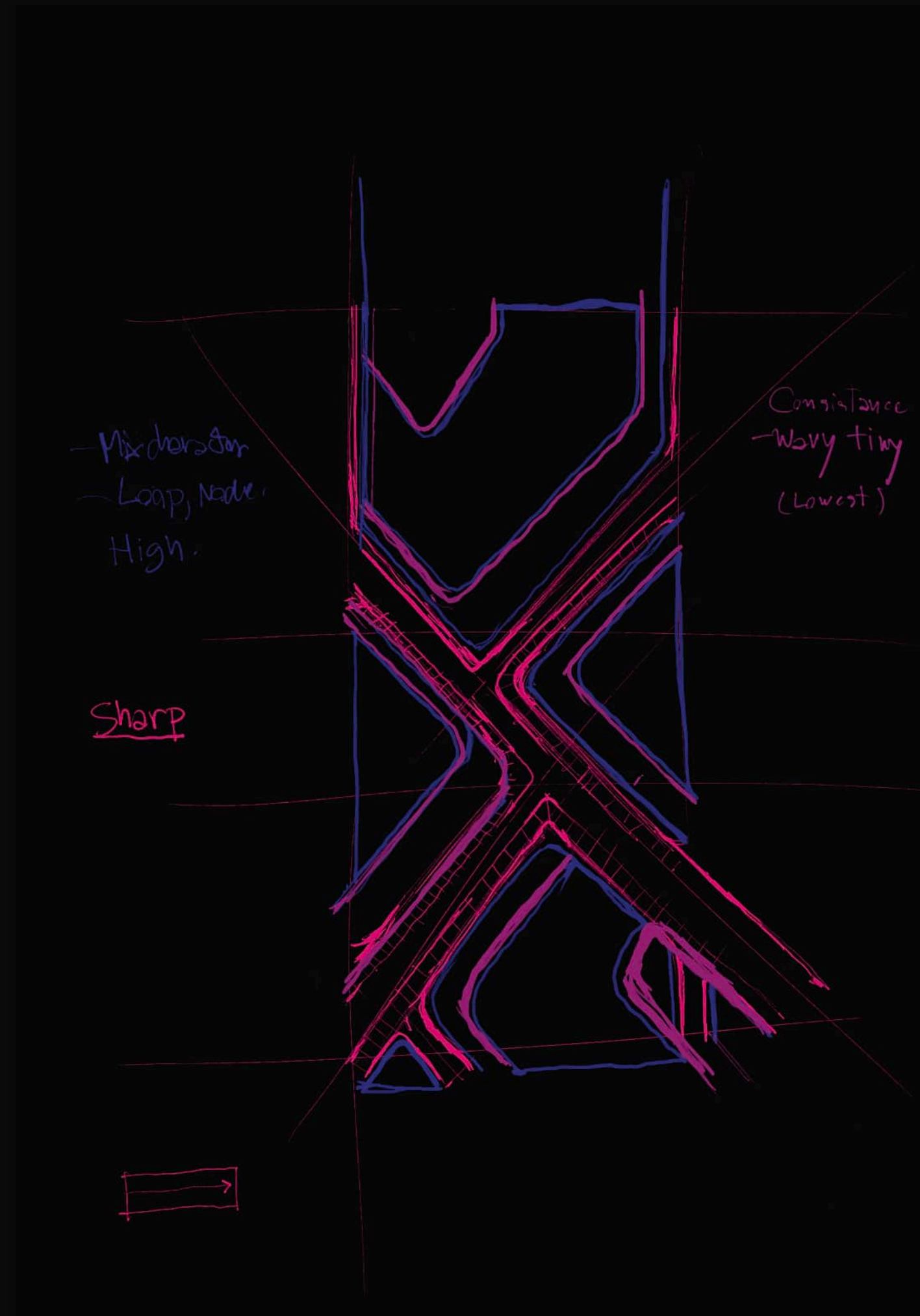
The background of the entire image is a close-up, slightly blurred photograph of a hand-drawn sketch on a textured, light-colored surface. A grey pen with an orange tip is visible in the lower right corner, having just finished drawing a line. The sketch consists of several parallel, slightly wavy lines that create a sense of depth and movement. The overall tone is muted and artistic.

FOLIO SUBMISSION

Kankawee Maksomboon | s4097770

SKETCHES

Design development process | Exprimental sketches design



FOLIO SUBMISSION

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REFLECTION

Reflection



Participating in the Tectonic Formation Lab as an Elective Based Research Assistant provided an intensive engagement with architectural fabrication rarely encountered within standard coursework. The hybrid GFRC, polymer, and metal facade prototype for Building 45 on Lygon Street demanded that research and making operate simultaneously — requiring technical precision and design intelligence at every stage.

The Tectonic Formation Lab elective provided two deeply interconnected learning experiences that together defined my understanding of digital fabrication in architectural practice.

The GFRC mould preparation pipeline was more demanding than anticipated. Each stage — plaster patching joint discontinuities, sequential sanding from 60 to 240-grit, resin sealing, UV curing, and crack repair — revealed how directly material decisions upstream affect casting quality downstream. Issues such as hairline cracks at foam joints, uneven resin application, and surface residue from cutting all required adaptive problem-solving on the workshop floor. These challenges reinforced that precision in preparation is inseparable from the quality of the final cast panel.

The digital design work across three Aurora iterations developed a working understanding of computational surface control — from Aurora-only exploration in Prototype_1, through behaviour-guided flow and ZBrush detail integration in Prototype_2, to fabrication-resolved developability in Prototype_3.

Marc Gibson's feedback at each iteration was instrumental — consistently pushing the geometry toward what was actually achievable in the mould. His Aurora tool demonstrated how computational design can generate genuinely complex architectural surfaces with material intelligence built in. Seeing that digital geometry realised as a physical GFRC prototype was a rare and rewarding confirmation that computational design and fabrication can operate as one continuous process.